

Technical Note TN-02: Permanent Magnet Architecture for an AF-MPD Thruster

Final Configuration for Continuous Operation, with Magnetic, Thermal and Discharge Analysis

Author: Dion Kuteesa | **Scope:** Theta Systems Applied-Field MPD Thruster | **Date:** September 2026

Abstract: This report defines the complete permanent magnet subsystem for a 1–5 kW breadboard applied-field magnetoplasmadynamic (AF-MPD) thruster running on argon propellant under **continuous steady-state operation**. It resolves three core engineering challenges. *Field Generation:* The segmented Halbach array delivers **0.631 T on-axis** at the cathode tip with magnets at their solved equilibrium operating temperature, exceeding the 0.40 T floor requirement with +57.7% margin. *Continuous Thermal Management:* An axial conduction architecture draws 500 W of plasma arc deposition backwards out of the copper anode through a full-diameter copper spreader and aluminium nitride (AlN) isolation barrier into an external cold plate, holding the NdFeB permanent magnets at **72 °C** (well below their 180 °C limit) with 488 W (97.6%) rejected through the rear face. *Discharge & Thrust Verification:* Whole-system plasma modeling and empirical scaling calibrated across 667 published argon firings predict a thrust of **186 mN at 100 A**. Total assembly mass is **7.17 kg** within a 114 mm diameter envelope.

1. Design Context & Objectives

This investigation establishes the magnetic and thermal architecture for a 1–5 kW class breadboard electric thruster operating on argon. Argon is roughly 1,000 times cheaper than xenon and completely free from noble gas supply vulnerabilities, but requires applied-field magnetoplasmadynamic acceleration to achieve high ionization efficiency and specific impulse ($I_{sp} > 2000 \text{ s}$).

Conventional AF-MPD thrusters rely on electromagnetic solenoids that impose severe mass, electrical power, and cooling burdens. This design replaces electromagnetic coils with an engineered permanent-magnet Halbach cylinder, eliminating electrical coil overhead and active magnet cooling loops.

Table 1: Primary Engineering Objectives and Achieved Values

Metric / Requirement	Specification Floor	Final Settled Design	Status
Cathode Tip Field (B_z) @ 0.40 T (hot)		0.631 T (hot, 72 °C) / 0.669 T (cold)	Passed (+57.7%)
Magnet Peak Temperature @ 100 T target / 180 T max		72 °C steady-state	Passed (-28 K)
Continuous Thermal Duty 500 W arc dissipation		488.2 W (97.6%) rear axial rejection	Passed
Total Assembly Mass @ 7.50 kg		7.17 kg (114 mm OD × 104 mm length)	Passed

2. Final Settled Design & Bill of Materials

The settled design is a 2D-axisymmetric assembly centered around a copper anode barrel and a central 2% thoriated tungsten cathode ($z=0$ defined at the mounting plate front face).

Table 2: Radial Assembly Build (Radius from Centerline)

Radius Range (r)	Component / Feature	Material	Engineering Function
0 to 3.175 mm	Cathode Rod (6.35 mm dia)	W, 2% thoriated	Thermionic arc attachment and emission
3.175 to 13.0 mm	Discharge Bore	Argon plasma	Annular propellant ionization & swirl channel
13.0 to 15.0 mm	Anode Barrel (2 mm wall)	Cu OFHC C101	Arc anode; axial heat conduction path
15.0 to 16.0 mm	Radial Thermal Break	Vacuum (1 mm gap)	Eliminates radial gas conduction
15.35 to 15.65 mm	Radiation Shield (0.3 mm)	304 SS, polished	Chokes radial thermal radiation ($\epsilon \approx 0.15$)
16.0 to 18.0 mm	Standoff Sleeve	Macor MGC Ceramic	Structural ceramic & electrical standoff
18.0 to 47.0 mm	Halbach Array (40 tiles)	Sintered NdFeB N48UH	Generates concentrated axial magnetic circuit
47.0 to 57.0 mm	Outer Return Casing (10 mm)	1010 / DC01 Low-C Steel	Magnetic flux containment and return path

Table 3: Halbach Cylinder Polarization Schedule

Ring Station	Axial Span (z)	Magnetisation Direction	Tile Count
Ring 1 (Rear Cap)	0 to 16 mm	Radially Outward ($\hat{r}$)	8 segmented tiles
Rings 2, 3, 4 (Core)	16 to 64 mm	Axial toward exhaust ($\hat{z}$)	24 segmented tiles
Ring 5 (Front Cap)	64 to 80 mm	Radially Inward ($-\hat{r}$)	8 segmented tiles

Table 4: Complete Assembly Bill of Materials (BOM)

Item	Description	Material	Qty	Mass (g)	Drawing Sheet
1	Mounting plate, hand-off boss	Cu OFHC C101	1	74.7	AFMPDT3-006
2	Mounting plate, magnet shelf	Macor MGC	1	229.2	AFMPDT3-006
3	Anode barrel, 2.0 mm wall	Cu OFHC C101	1	78.8	AFMPDT3-002
4	Radiation shield, 0.3 mm wall	304 SS, polished	1	11.7	AFMPDT3-005
5	Standoff sleeve	Macor MGC	1	43.1	AFMPDT3-005
6	Magnet tile, rear cap, RADIAL OUT	NdFeB N48UH	8	710.6	AFMPDT3-003
7	Magnet tile, core, AXIAL +z	NdFeB N48UH	24	2131.9	AFMPDT3-003
8	Magnet tile, front cap, RADIAL IN	NdFeB N48UH	8	710.6	AFMPDT3-003
9	Iron return casing, 10 mm wall	1010 / DC01 steel	1	2047.4	AFMPDT3-004
10	Front retaining clamp ring	304 SS	1	425.0	AFMPDT3-009
11	Spreader, full diameter	Cu OFHC C101	1	531.0	AFMPDT3-007
12	Isolation plate, dielectric	AlN, sintered	1	65.5	AFMPDT3-008
13	Cathode rod, 6.35 mm dia	W, 2% ThO2	1	58.7	AFMPDT3-010
14	Cathode insulating bush	Macor MGC	1	1.3	AFMPDT3-010
15	Screw insulating bushes	Alumina 99.7%	6	2.4	AFMPDT3-007
16-18	Fasteners & flat washers	A2-70 Stainless	24	49.2	Standard

3. Magnetic Field Performance & Verification

The centerline axial field $B_z(z)$ reaches 0.669 T cold and 0.631 T at full steady-state operating temperature (72 °C), calculated using the N48UH reversible thermal coefficient $\alpha_{Br} = -0.11\%/ \text{K}$.

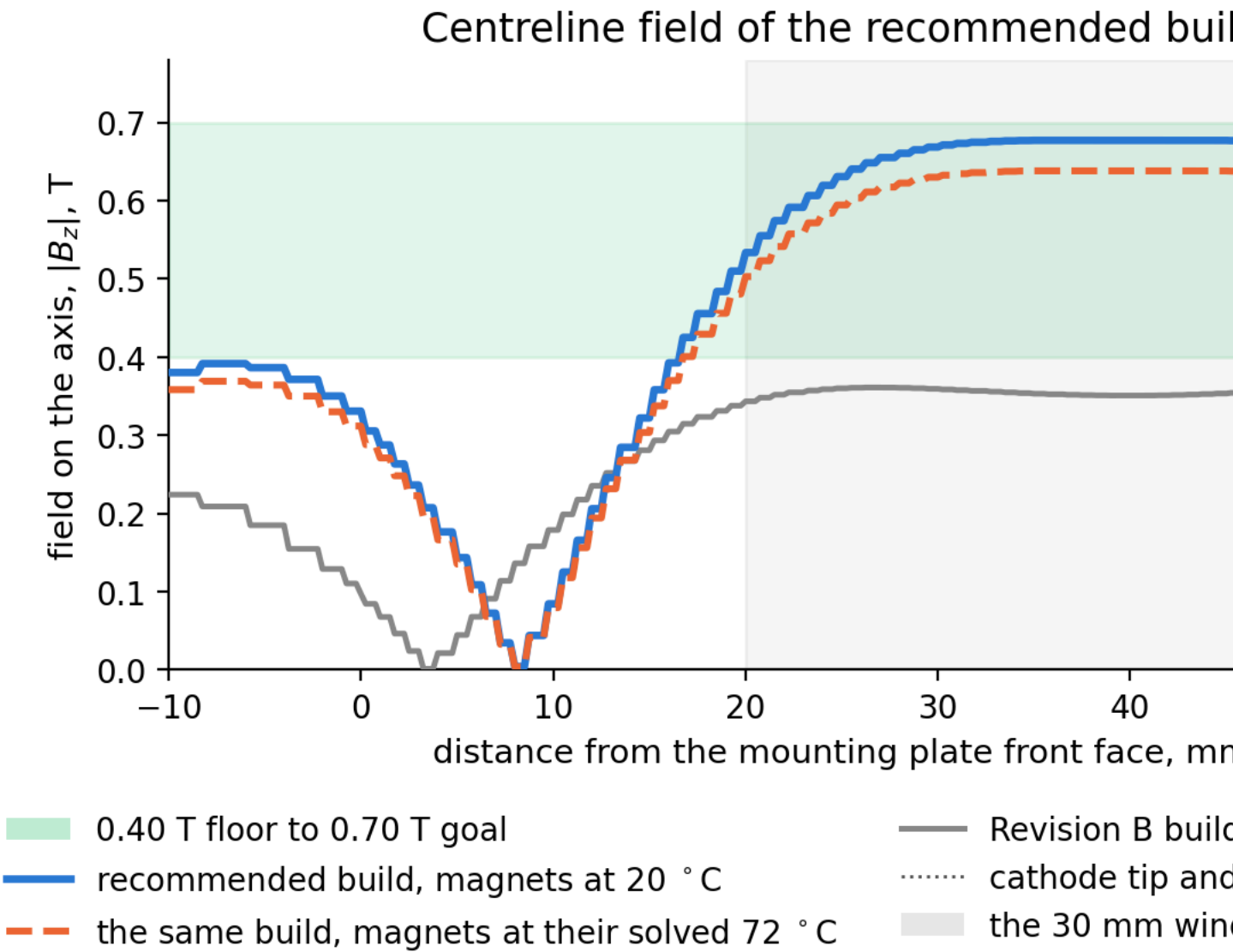
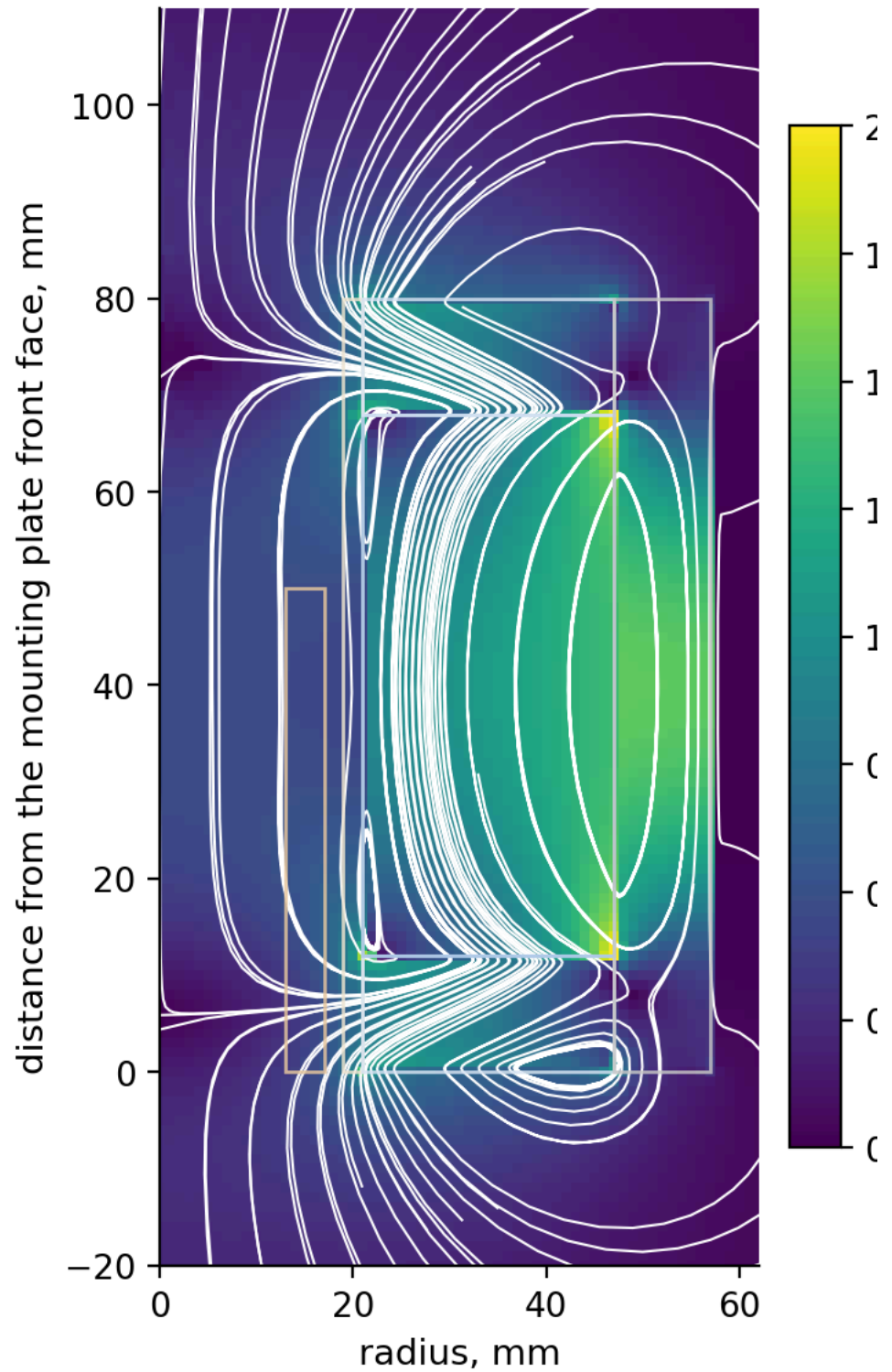


Figure 1: Centerline magnetic field $B_z(z)$ cold (20 °C) and hot (72 °C steady state), maintaining $B_z > 0.50 \text{ T}$ across the entire discharge channel.

Flux path through the assembly



Nearest exported build: 12 mm end caps, 94 mm magnet OD. The flux path is the same for

Figure 2: 2D axisymmetric flux topology showing axial magnetic flux channeled through the outer steel casing and crossing the bore through the radial end caps.

Two independent codes on identical geometri

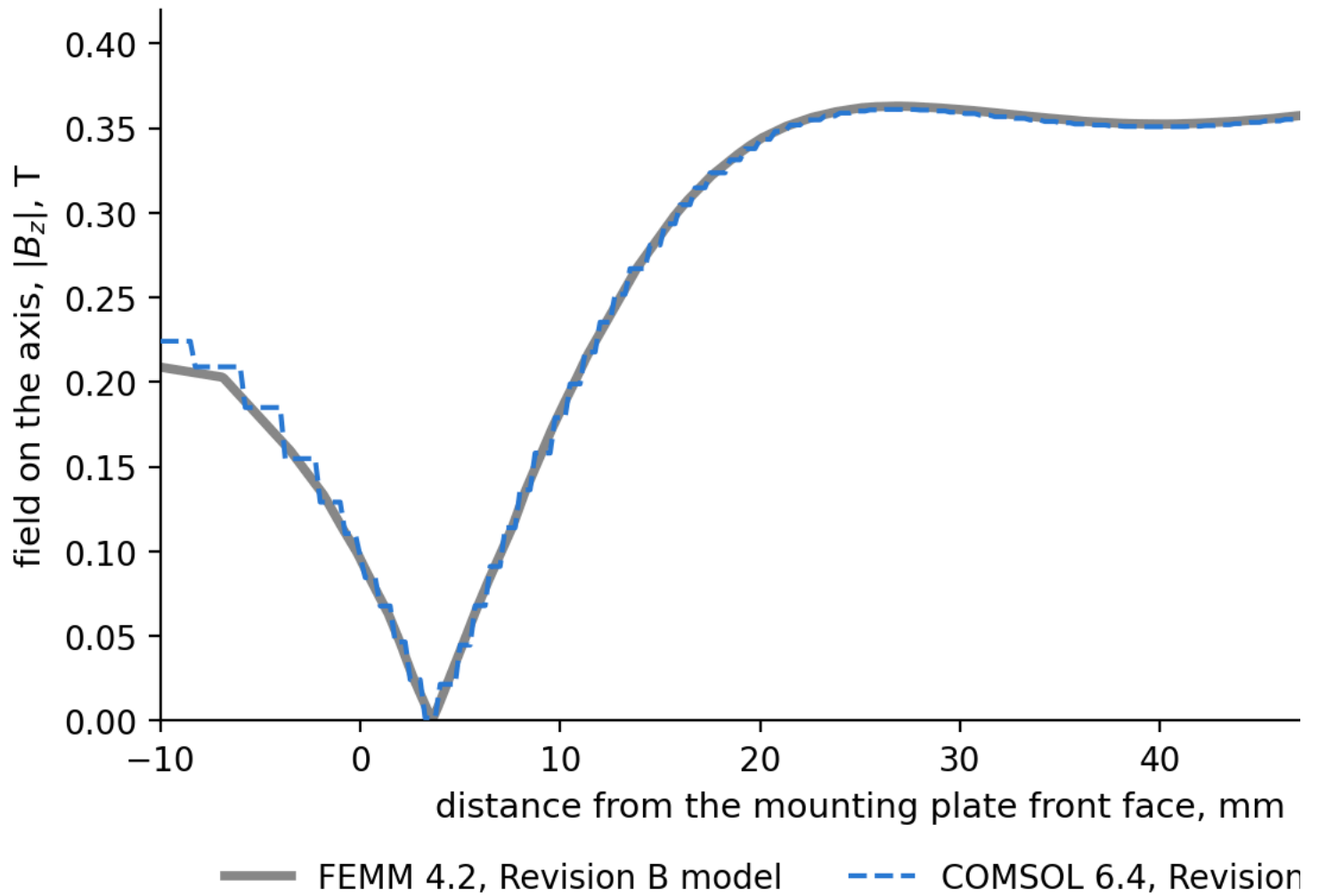


Figure 3: Independent cross-code validation between FEMM 4.2 and COMSOL Multiphysics 6.4, matching within 0.02% across the acceleration bore.

Demagnetisation margin of the recommen

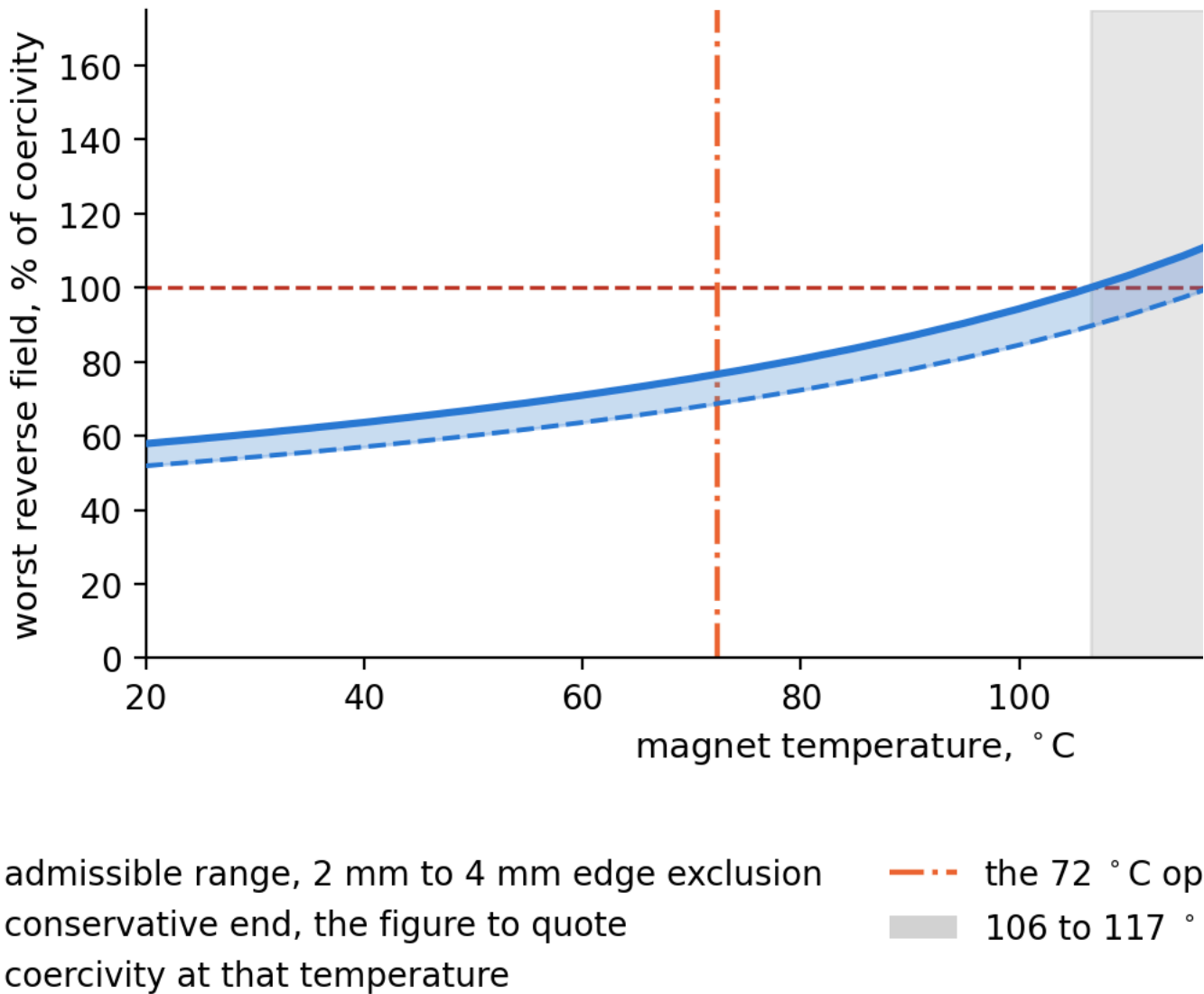
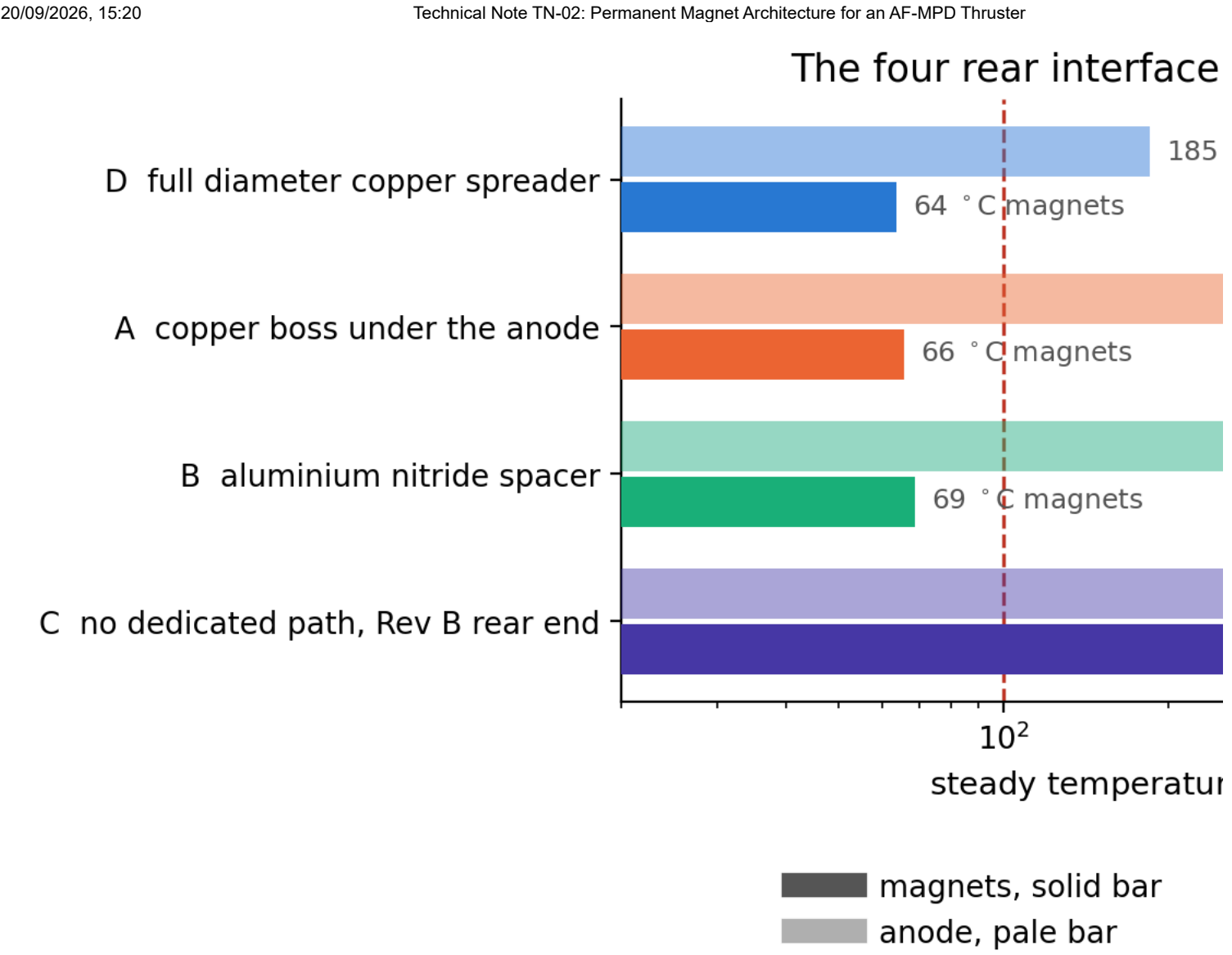


Figure 4: Operating reverse field vs intrinsic coercivity $H_{cj}(T)$, demonstrating at least a 23% safety margin against demagnetization at 72 °C.

4. Continuous Thermal Management Architecture

The thermal architecture resolves the continuous 500 W arc deposition via an engineered rear conduction route (Concept D), spreading the heat over the entire 114 mm diameter rear chassis.

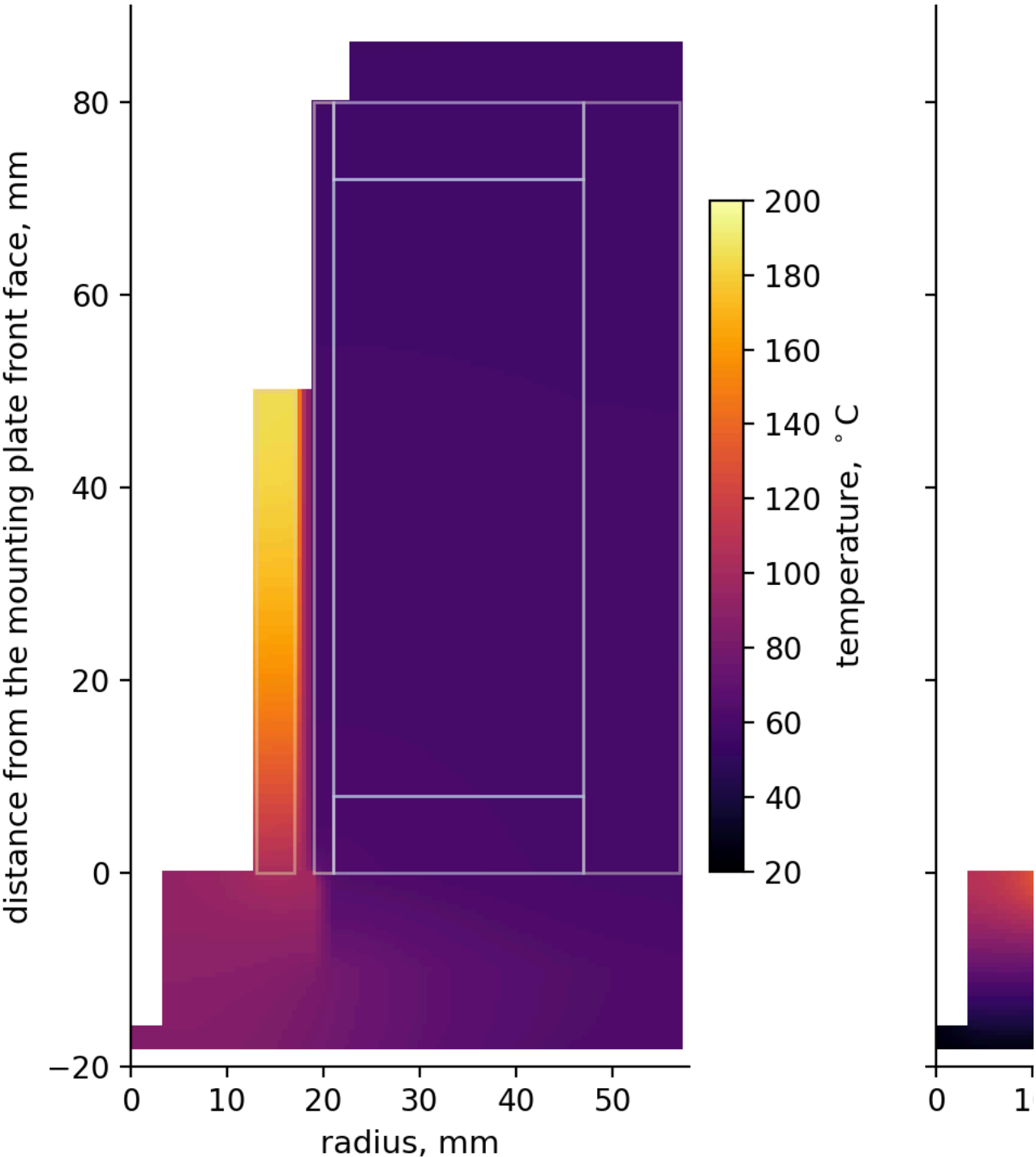


Steady temperature at 500 W continuous. No

Concept D, copper spreader

peak 185 °C

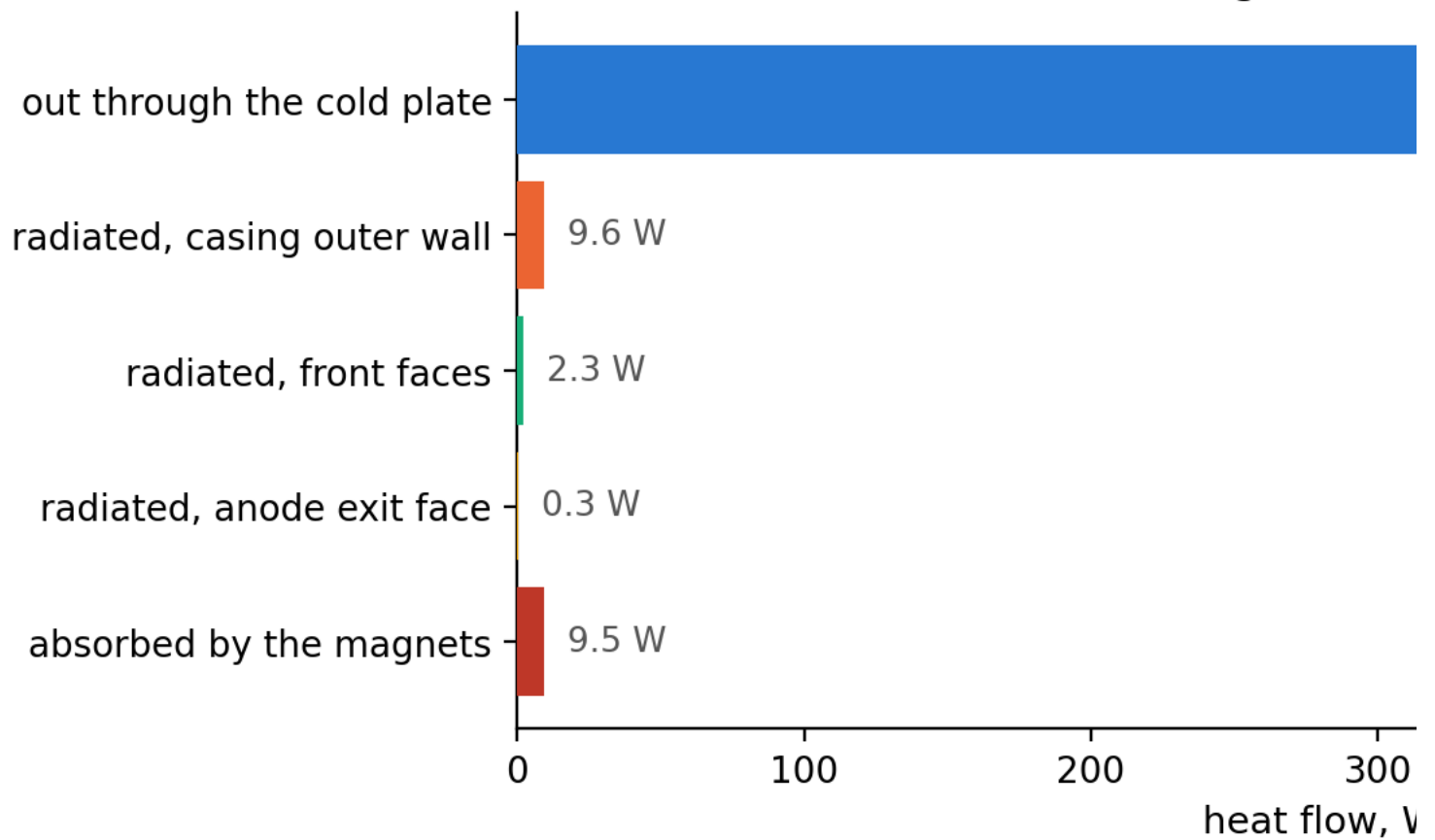
Con



Nearest exported build: 4 mm anode wall, 2 mm break. The recommended build runs 8 K h

Figure 6: 2D equilibrium temperature distribution at 500 W continuous operation ($Sh_{\text{interface}} = 1000 \text{ ext{ W/m}^2\cdot\text{ext{K}}}$).

Where the 500 W goes, re



The magnet term is what leaks in, not a way out

Figure 7: Steady-state thermal budget: 488.2 W (97.6%) conducted to external cold plate, choking radial leak into magnet bore to $< 4 \text{ W}$.

The magnet bore is expensive in field and the
what the bore radius costs

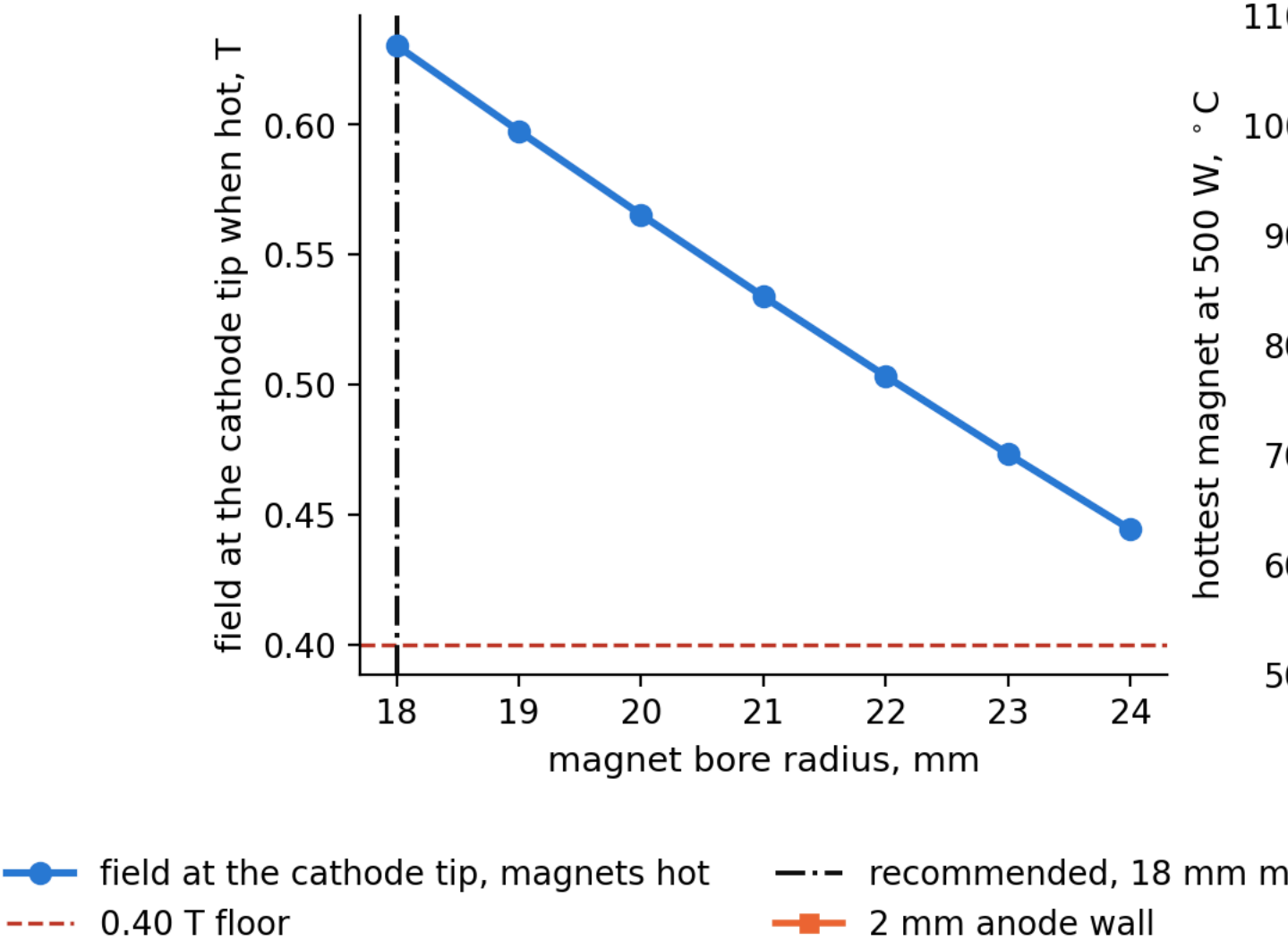


Figure 8: Geometry trade: on-axis field degradation per mm bore expansion vs magnet temperature as a function of thermal break width.

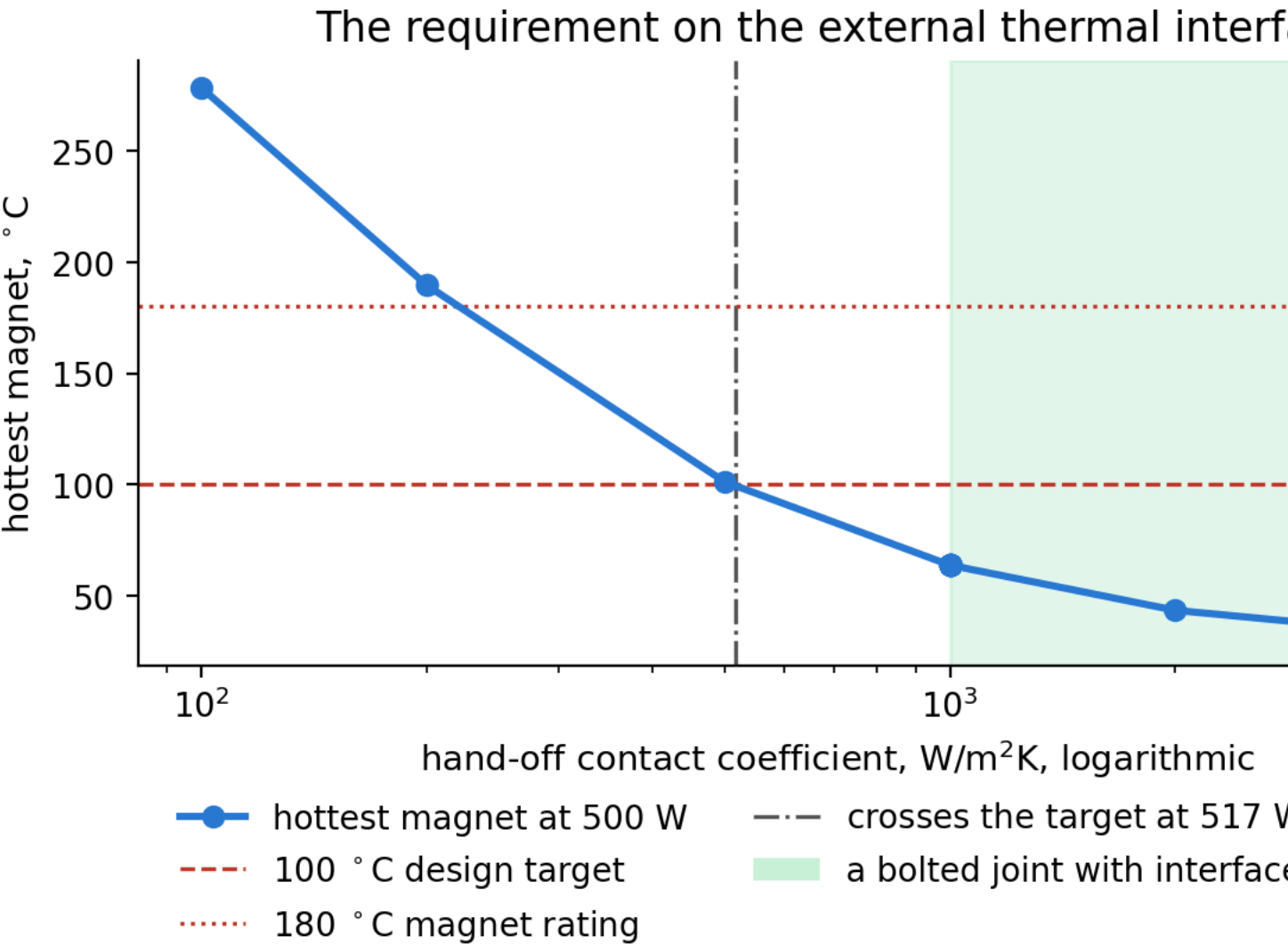


Figure 9: Peak magnet temperature vs cold plate interface conductance $h_{\text{interface}}$. The 100 $^{\circ}C$ target is met for conductances $>520 \text{ W/m}^2 \cdot \text{K}$.

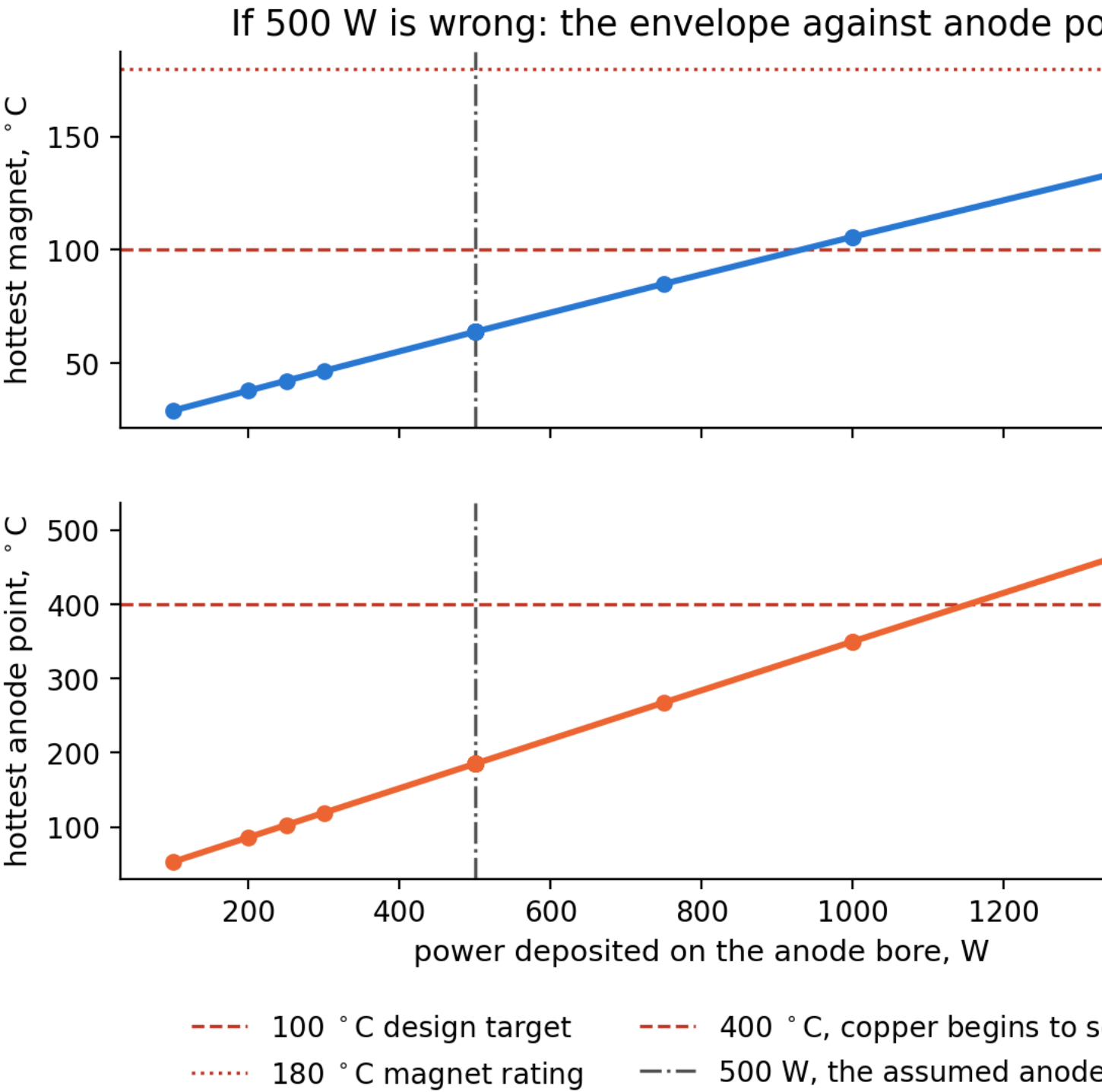
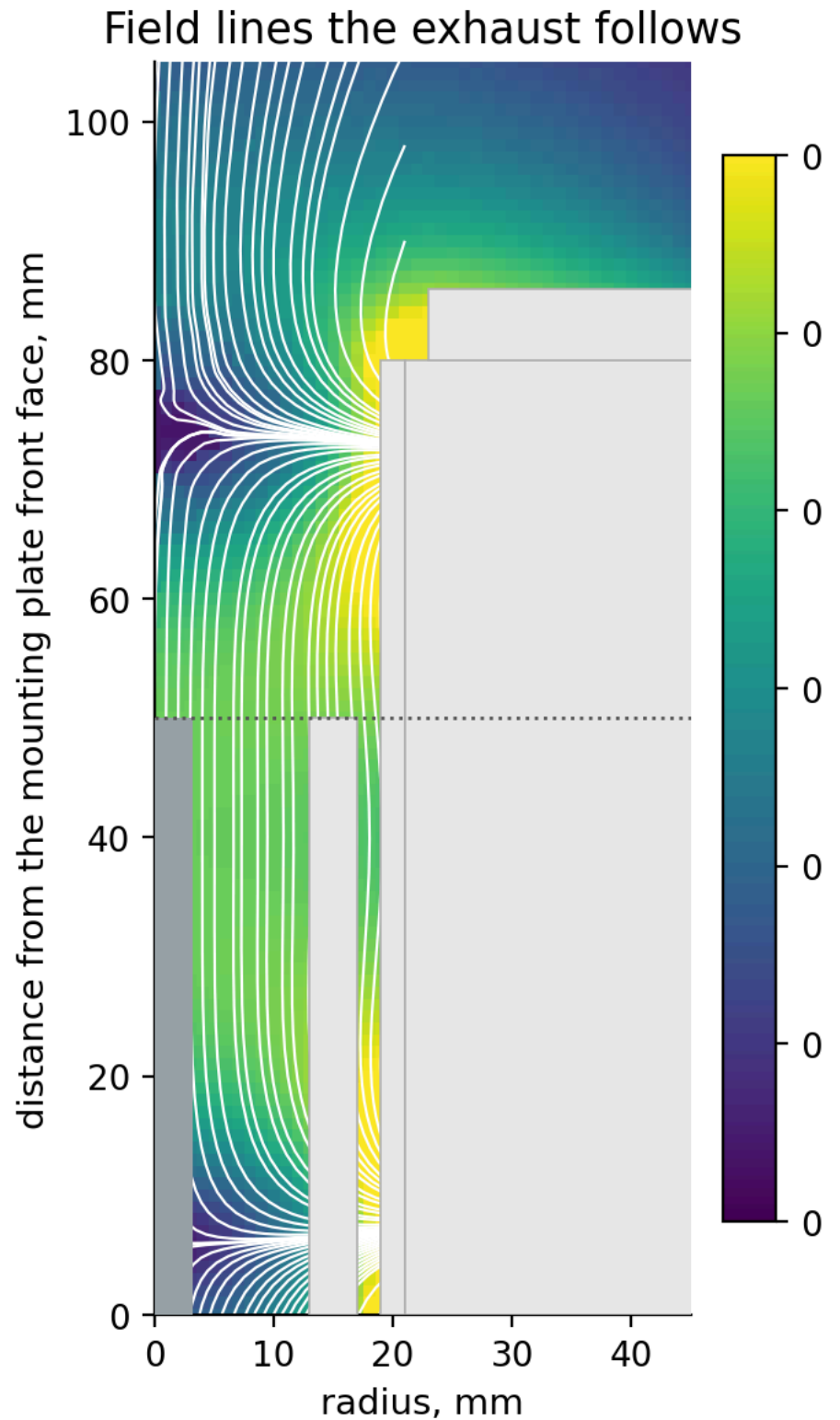


Figure 10: Thermal envelope scaling up to 1500 W continuous input; magnets remain below 100 °C target up to 760 W continuous arc deposition.

5. Whole System & Discharge Modeling

Downstream of the cathode tip, diverging magnetic field lines form a magnetic nozzle that expands swirling plasma and converts azimuthal swirl into directed axial kinetic energy.



*Nearest exported build: 12 mm end caps. Hardware is drawn solid; the dotted line is the ca
The axis mesh coarsens past 60 mm, so the plume is indicative rather than resolve*

Figure 11: Magnetic nozzle field line tracing into the plume. The active discharge channel maintains field strengths $>0.10 \text{ T}$ over 98% of its volume.

The discharge, and where it accelerates the

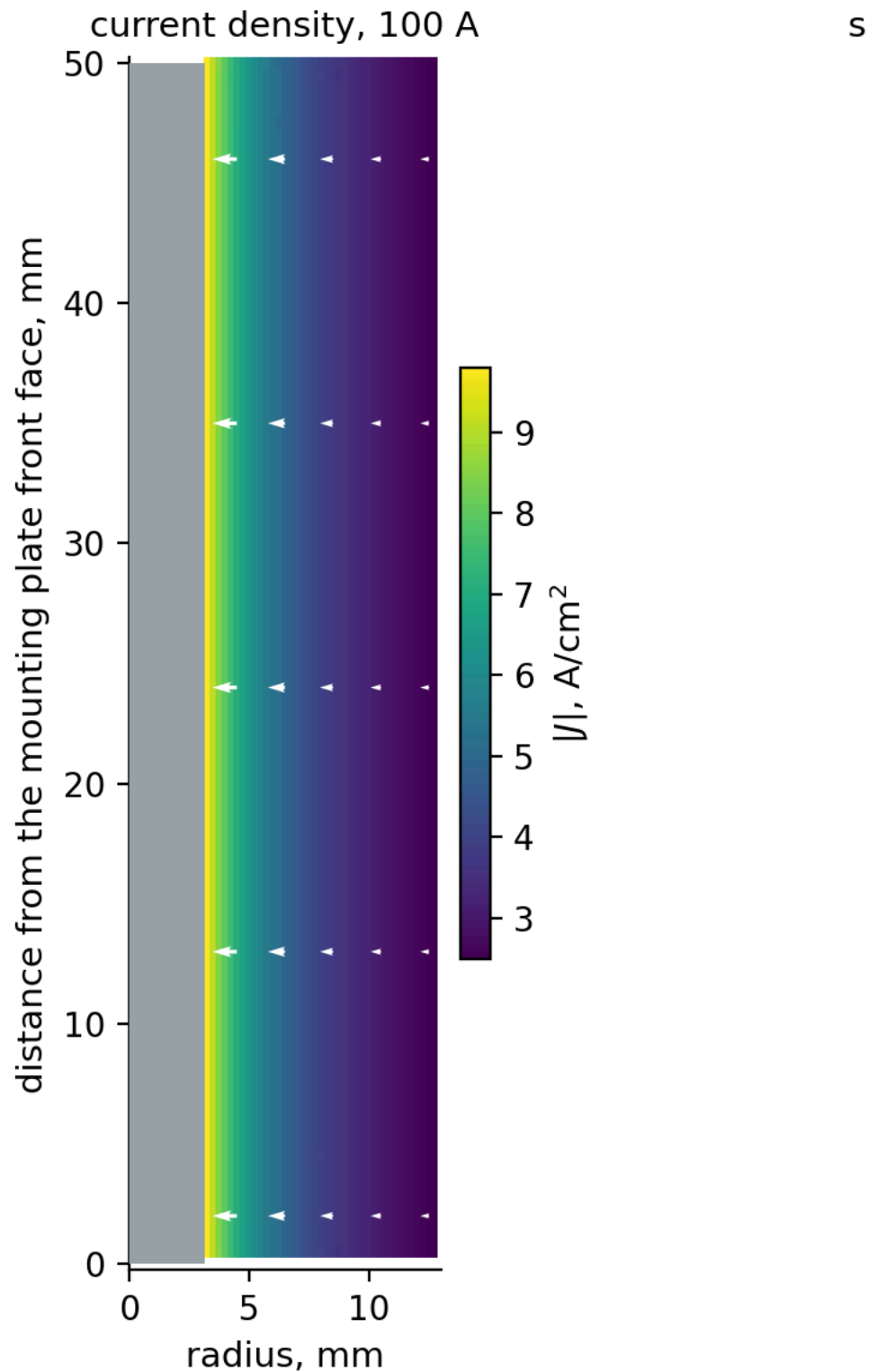
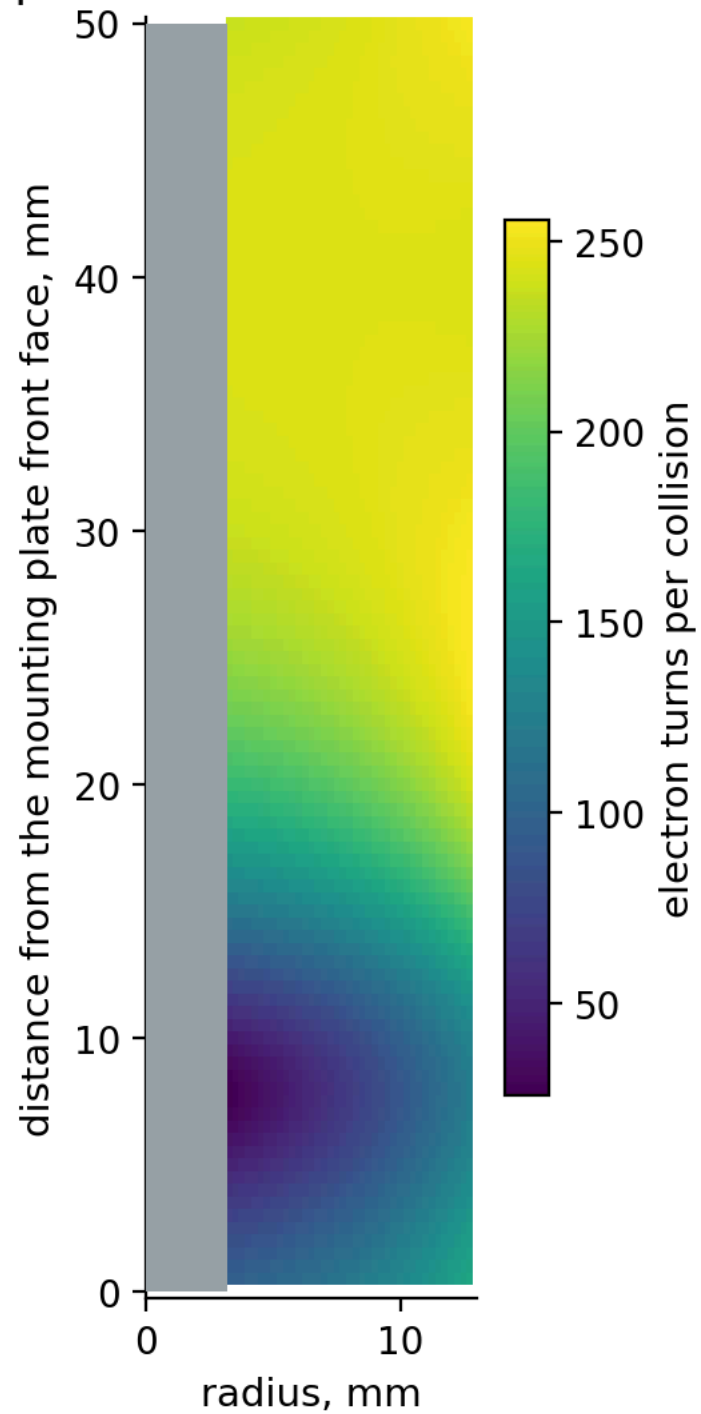


Figure 12: Arc current streamline paths and azimuthal Lorentz body force density $f_{\theta} = j_r B_z$ driving propellant swirl acceleration.

Hall parameter across the channel



Median 235 across the channel. Above about one the conductivity is a tensor, so read the current paths as shape rather than as exact.

Figure 13: Electron Hall parameter (η_e) contours throughout the discharge channel, indicating a median value of 235.

The feasible set: field, flatness and mass ca

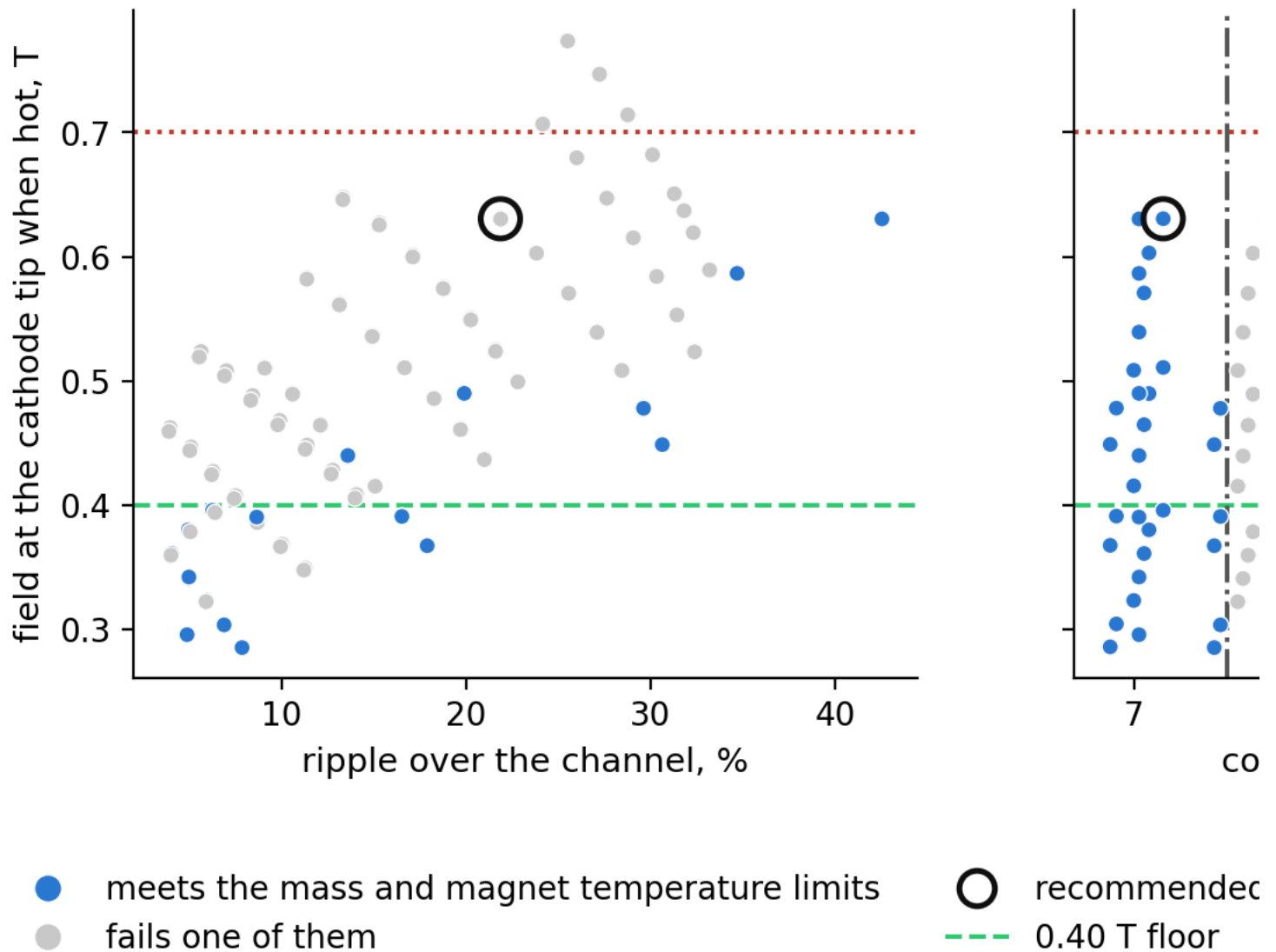


Figure 14: Mass vs operating field for 136 magnetostatic builds. The selected 7.17 kg build (circled) represents the global optimum within the 7.5 kg envelope.

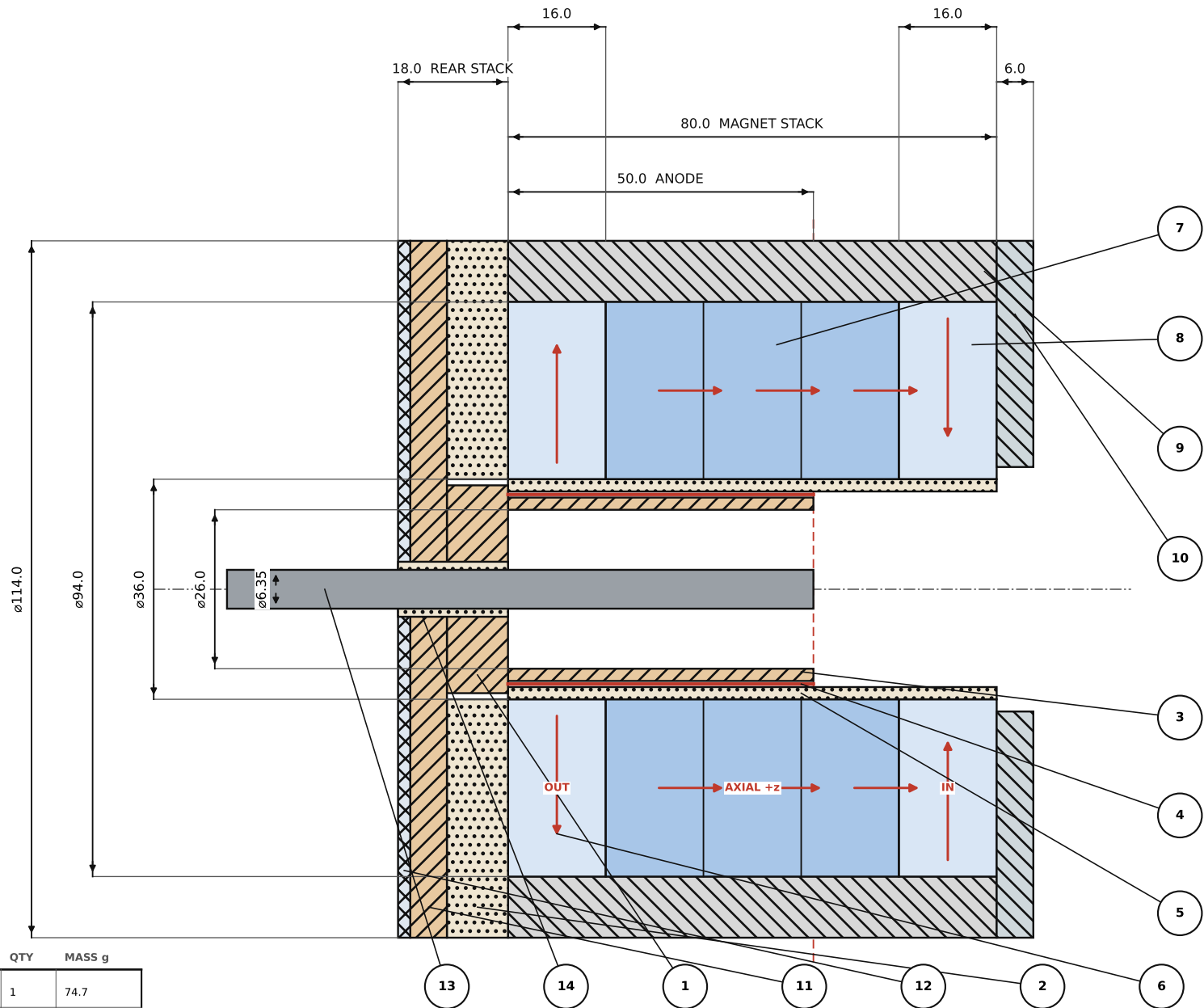
Predicted Thrust Performance: Calibrating empirical scaling against the 667-run published argon AF-MPD database predicts a thrust of **186 mN at 100 A** (likely range: 93 to 226 mN).

6. End Cap Configuration Trade

The baseline build uses 16 mm end caps to maximize tip field (0.631 T hot over 31.5 mm). An alternative 12 mm end cap build meets both field floor and shape requirements simultaneously (0.511 T hot over 40.8 mm) at the exact same 7.16 kg mass and identical hardware envelope.

7. Manufacturing Drawing Package

A complete 10-sheet A3 manufacturing drawing set is appended to this report, incorporating full GD&T, material certifications, fastener preloads (ISO 4762 M4 at 1.5 N·m), and magnetization schedules.



PARTS LIST

ITEM	DESCRIPTION	MATERIAL	QTY	MASS g
1	Mounting plate, hand-off boss	Cu OFHC C101	1	74.7
2	Mounting plate, magnet shelf	Macor MGC	1	229.2
3	Anode barrel, 2.0 mm wall	Cu OFHC C101	1	78.8
4	Radiation shield, 0.3 mm wall	304 SS, polished	1	11.7
5	Standoff sleeve	Macor MGC	1	43.1
6	Magnet tile, rear cap, RADIAL OUT	NdFeB N48UH	8	710.6
7	Magnet tile, core, AXIAL +z	NdFeB N48UH	24	2131.9
8	Magnet tile, front cap, RADIAL IN	NdFeB N48UH	8	710.6
9	Iron return casing, 10 mm wall	1010 / DC01 steel	1	2047.4
10	Front retaining ring	304 SS	1	425.0
11	Spreader, full diameter	Cu OFHC C101	1	531.0
12	Isolation plate	AlN, sintered	1	65.5
13	Cathode rod, 6.35 dia	W, 2 % ThO2	1	58.7
14	Cathode insulating bush	Macor MGC	1	1.3
15	Screw insulating bush	Alumina 99.7 %	6	2.4
16	ISO 4762 M4 x 30 socket head	A2-70	6	28.2
17	ISO 4762 M4 x 16 socket head	A2-70	6	16.8
18	ISO 7089 4 mm flat washer	A2	12	4.2
19	Feed tube stub, 1/8 in x 0.7 wall, brazed	Cu	1	3.0
	COMPLETE ASSEMBLY			7174

NOTES

- Datum z = 0 is the front face of the mounting plate. Exhaust is +z, to the right. The assembly runs z = -18 to z = +86 and is 114 mm across.
- The cathode tip sits at z = +50, flush with the anode exit, on the dashed line. That is where the applied field is defined and the field falls steeply behind it.
- Magnetisation direction is shown by the red arrows and is specified per ring on AFMPDT3-003. It is set at manufacture and cannot be changed afterwards.
- The rear face of item 12, at z = -18, is the thermal hand-off face. It mounts to an external cold plate and must be flat within 0.05 over the full diameter.
- The 1.0 radial gap between items 1 and 2, and the 1.0 evacuated break between the anode and the sleeve, must remain clear and evacuated. Both are thermal breaks. The 0.3 radiation shield, item 4, floats in the second of these.
- Items 16 to 18 pass through insulating bushes, item 15, in the spreader, so that the cold plate stays isolated from the anode.
- Complete assembly mass 7.17 kg.

GENERAL ARRANGEMENT - RECOMMENDED BUILD

AF-MPDT Breadboard / Rev 3

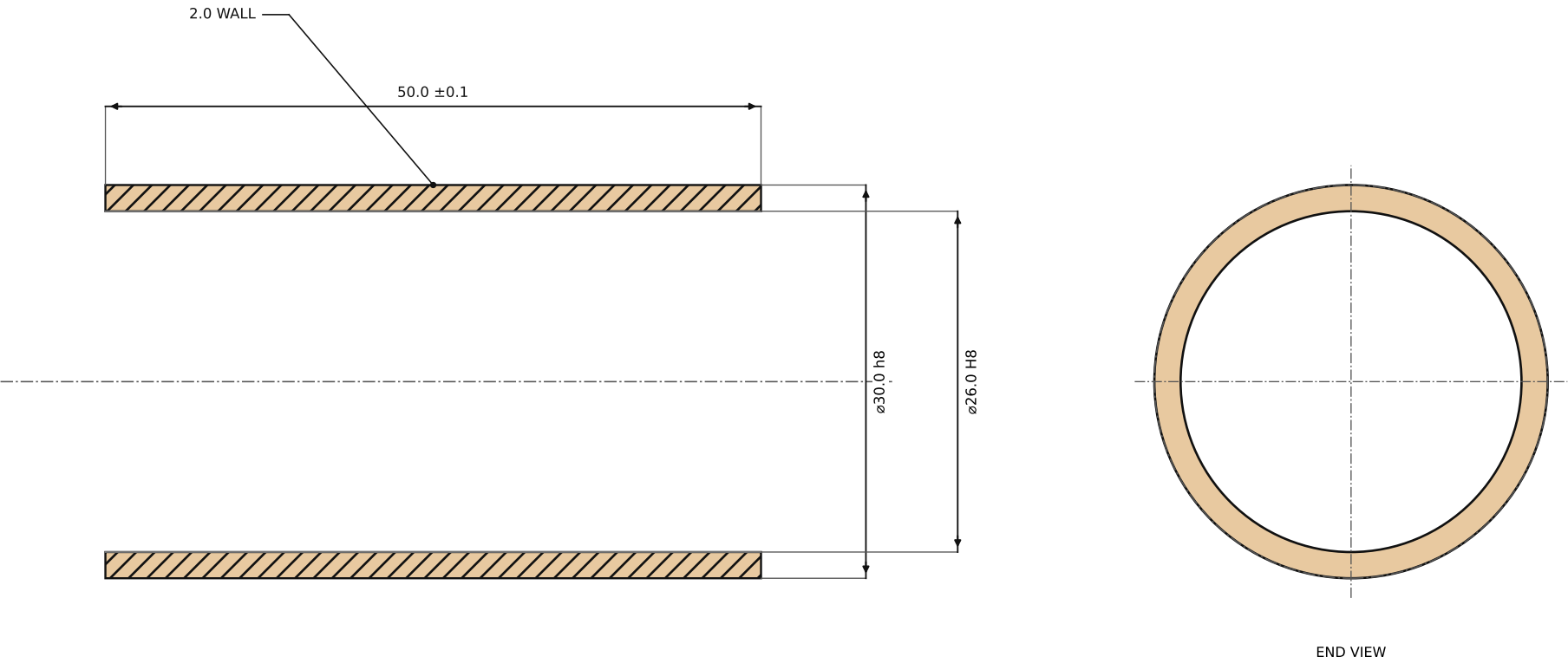
DRG No.
AFMPDT3-001

MATERIAL
ASSEMBLY

SCALE
1:1

DATE
2026-09-12

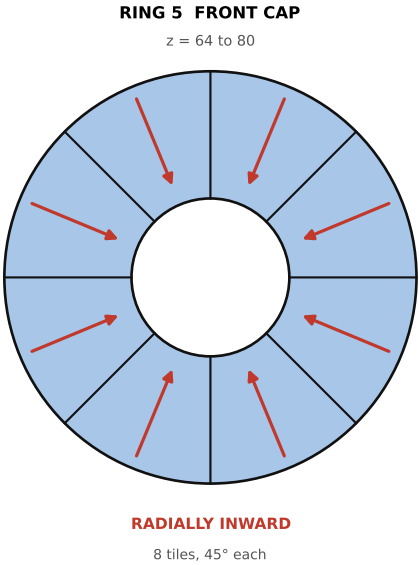
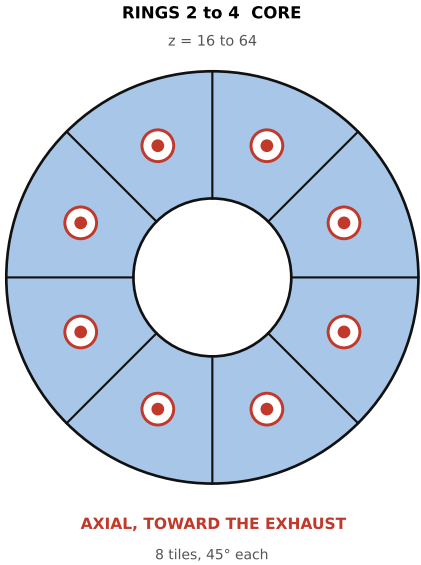
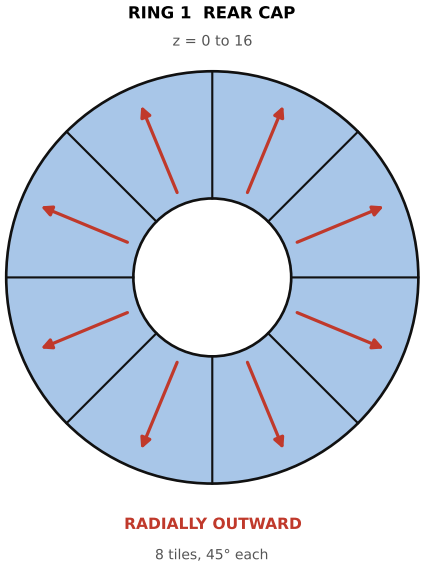
ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED.
REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.



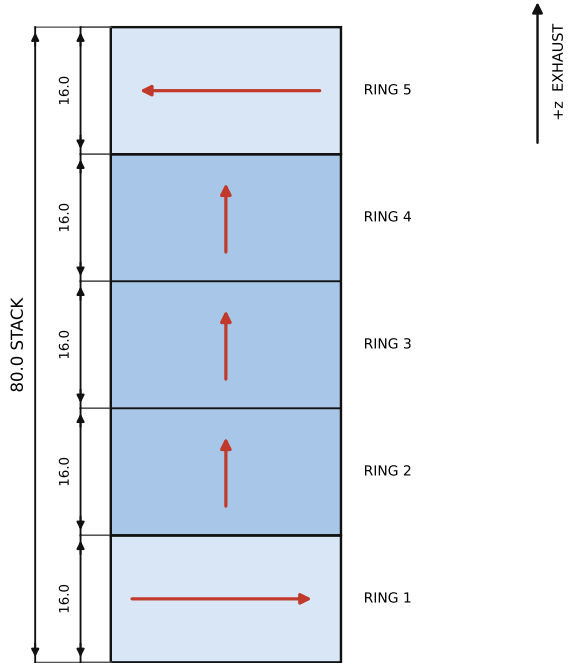
- NOTES**
- 1. Cu OFHC C101, half hard. Do not anneal after final machining.
 - 2. Bore Ra 0.8 max. Outside diameter Ra 1.6 max.
 - 3. Both ends square to the bore within 0.02.
 - 4. Rear face flat within 0.02: it is the conduction joint to the hand-off boss and carries 500 W in service.
 - 5. No rear flange. The barrel is carried and cooled by the mounting plate, AFMPDT3-006.
 - 6. Emissivity of the outer surface is taken as 0.30 in the thermal model. Do not polish the outside.
 - 7. Mass 78.8 g.

ANODE BARREL		
AF-MPDT Breadboard / Rev 3		DRG No. AFMPDT3-002
MATERIAL Cu OFHC C101	SCALE 2:1	DATE 2026-09-12
ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED. REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.		

ALL THREE VIEWS LOOK UPSTREAM, AGAINST THE EXHAUST. ⊙ IS MAGNETISATION OUT OF THE PAGE, TOWARD THE VIEWER, WHICH IS THE EXHAUST DIRECTION.

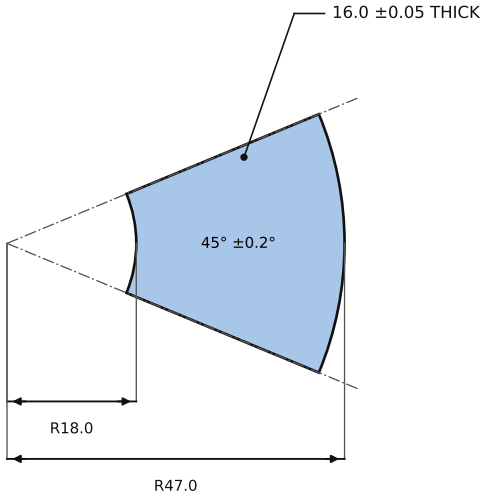


STACK SECTION



ONE TILE

40 off, all identical in shape



MAGNETISATION SCHEDULE

RING	REGION	z mm	MAGNETISATION	QTY	PART SUFFIX
1	REAR CAP	0 to 16	RADIAL OUT	8	-R
2	CORE	16 to 32	AXIAL +z	8	-A
3	CORE	32 to 48	AXIAL +z	8	-A
4	CORE	48 to 64	AXIAL +z	8	-A
5	FRONT CAP	64 to 80	RADIAL IN	8	-F

NOTES

- Sintered NdFeB grade N48UH. Br 1.38 to 1.42 T and intrinsic coercivity H_{cj} not less than 1990 kA/m, both at 20 °C. Maximum working temperature 180 °C.
- Each tile is magnetised through its own mid plane in the direction given by the schedule. Radial tiles are magnetised along the radial bisector of the 45° segment.
- MAGNETISATION DIRECTION IS SET AT MANUFACTURE AND CANNOT BE CHANGED AFTERWARDS. It must appear on the purchase order per ring, not only the grade and the geometry.
- Built with rings 1 and 5 the other way round, the field at the cathode tip falls from 0.669 T to 0.072 T, a factor of five.
- Mark each tile with its part suffix and a direction arrow.
- Coating NiCuNi, 10 to 25 um. Chamfer 0.3 x 45° all edges.
- Tolerance on radii ±0.1, on thickness ±0.05, on the included angle ±0.2°.
- Mass 88.8 g per tile, 3553 g for the stack of 40.

MAGNET TILES AND MAGNETISATION SCHEDULE

AF-MPDT Breadboard / Rev 3

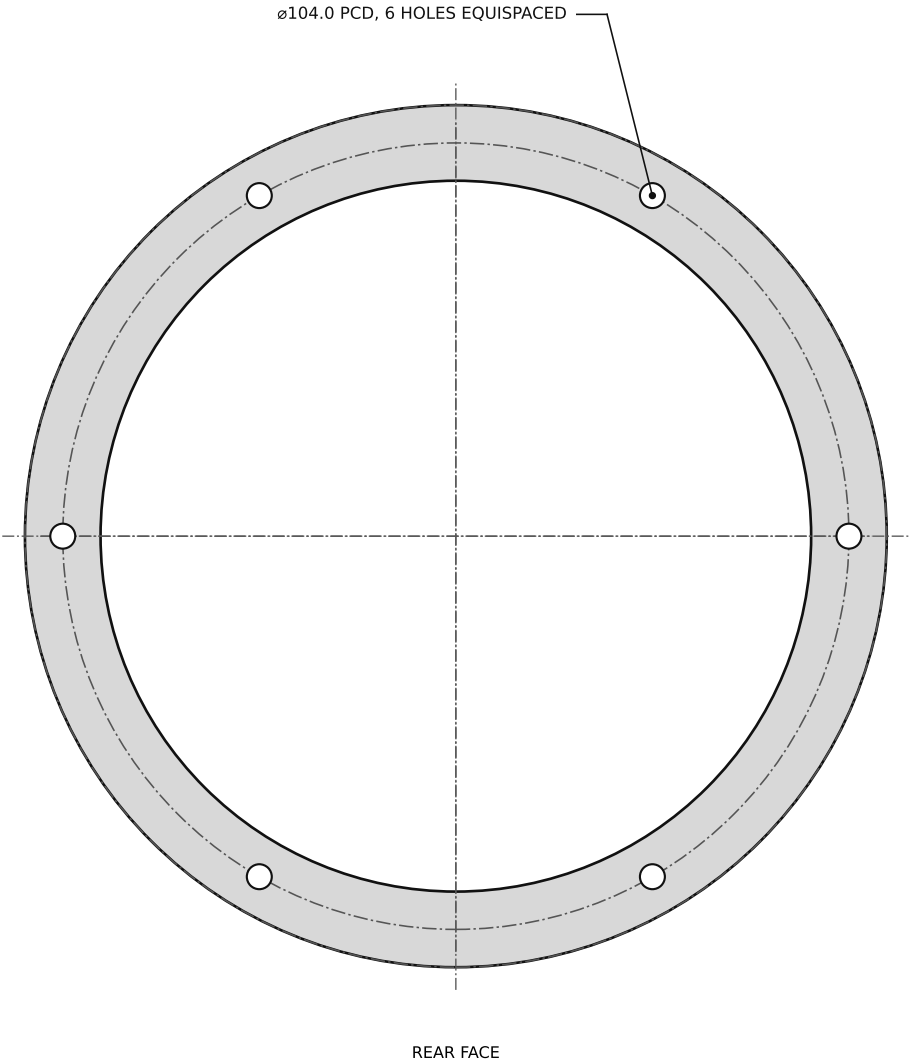
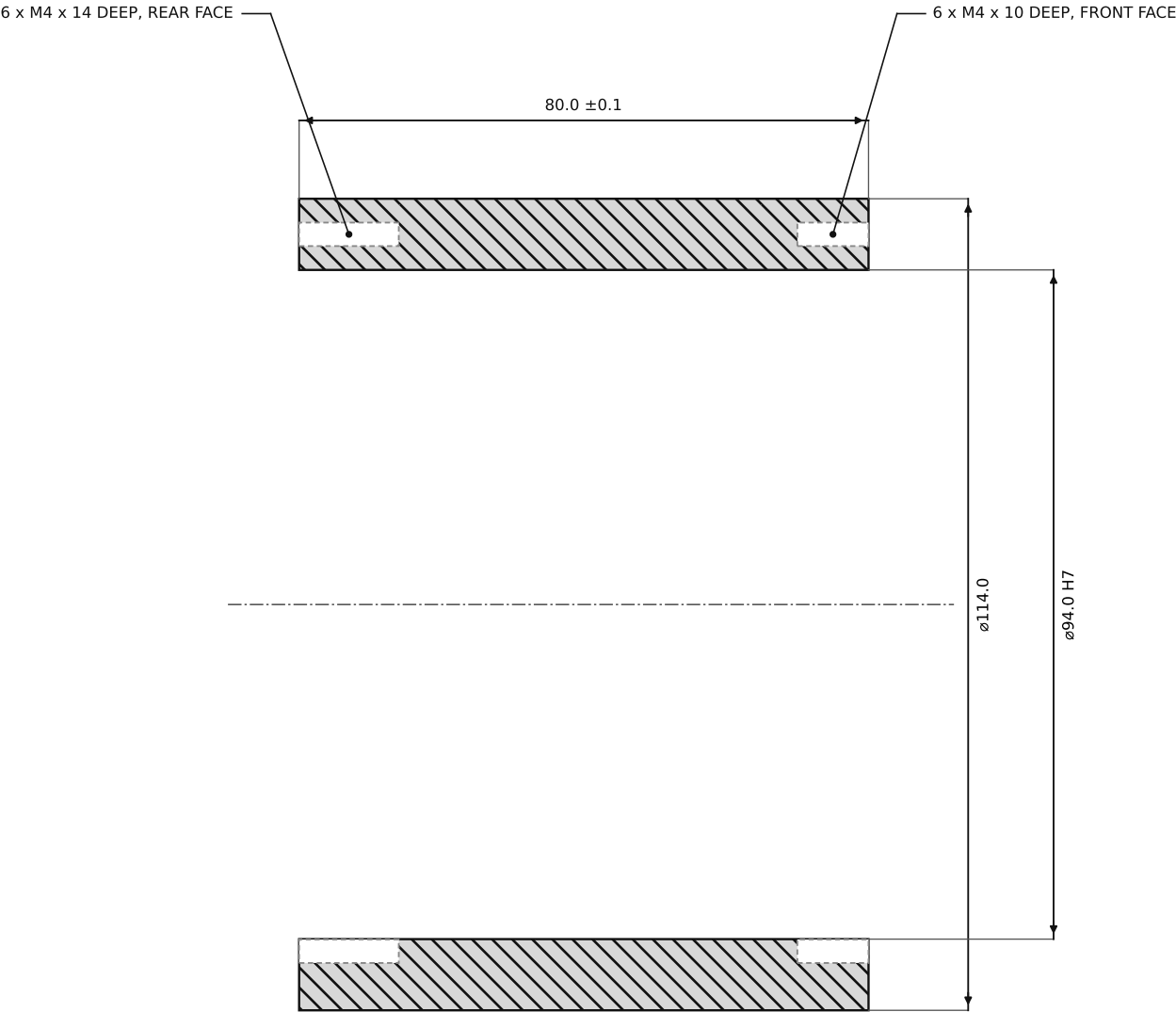
DRG No.
AFMPDT3-003

MATERIAL
NdFeB N48UH

SCALE
1:1 / 1:2

DATE
2026-09-12

ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED.
REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.



- NOTES**
1. Must be a FERRITIC grade. Austenitic stainless is non magnetic and would delete the flux return path entirely.
 2. Fully annealed after machining, to restore permeability.
 3. Bore ø94.0 H7, concentric with the outside within 0.05.
 4. Both end faces square to the bore within 0.05 and flat within 0.05. They carry the clamp load.
 5. Outer surface blackened or oxide coated, emissivity not less than 0.9. It radiates about 10 W in service.
 6. Peak flux density in service is 1.58 T away from the magnetisation corners, against a saturation knee near 2.1 T.
 7. Mass 2047 g.

IRON RETURN CASING

AF-MPDT Breadboard / Rev 3

DRG No.
AFMPDT3-004

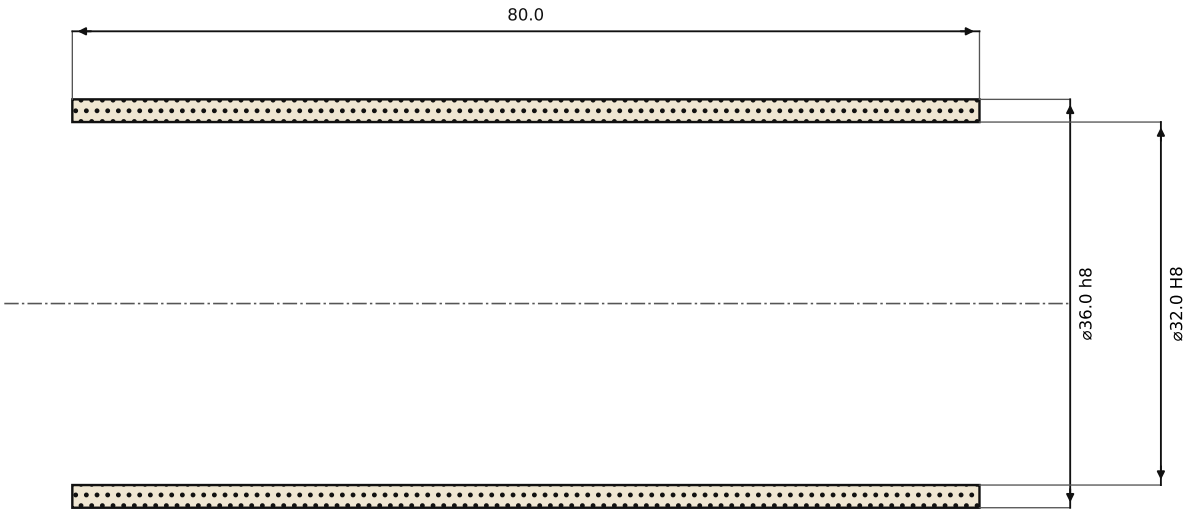
MATERIAL
1010 / DC01 low carbon steel

SCALE
1:1

DATE
2026-09-12

ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED.
REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.

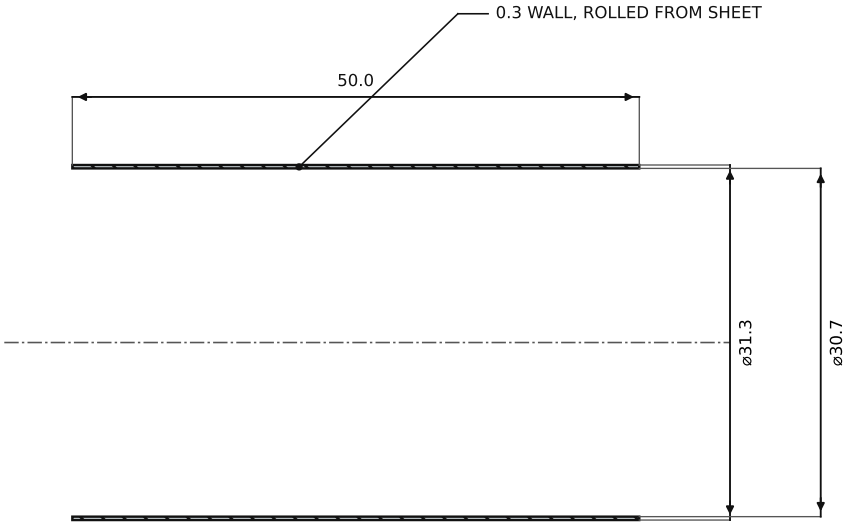
ITEM 5 STANDOFF SLEEVE, Macor MGC



NOTES

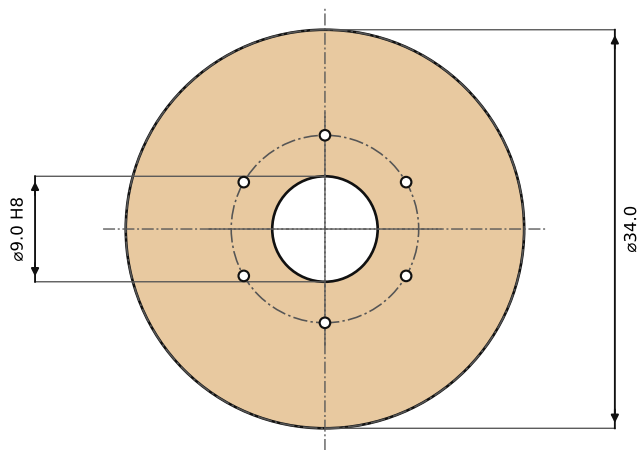
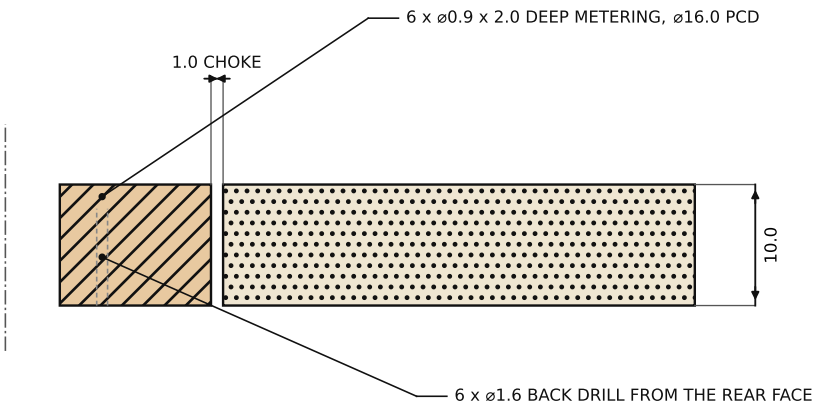
- SLEEVE, item 5
1. Macor MGC machinable glass ceramic. Bake at 200 °C for 4 hours under vacuum before first use.
 2. It locates the magnet bore and carries no heat of consequence: the thermal model gives 0.17 W entering it at 500 W.
 3. Concentricity of bore to outside within 0.05.
- SHIELD, item 4
4. Rolled from 0.3 mm 304 sheet, seam left open. It carries no load and floats in the 1.0 mm evacuated break.
 5. BOTH FACES POLISHED to Ra 0.2 or better, target emissivity 0.05. Handle with gloves only; fingerprints raise emissivity.
 6. Radial clearance is 0.35 each side. The anode grows 0.07 on radius at its 288 °C service temperature, so the gap does not close.
 7. The shield is worth about 1 K on magnet temperature. It may be deleted without consequence if the 0.35 clearance is awkward.
8. Masses: sleeve 43.1 g, shield 11.7 g.

ITEM 4 RADIATION SHIELD, 304 SS



STANDOFF SLEEVE AND RADIATION SHIELD		
AF-MPDT Breadboard / Rev 3		DRG No. AFMPDT3-005
MATERIAL Macor MGC / 304 SS	SCALE 1.5:1	DATE 2026-09-12
ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED. REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.		

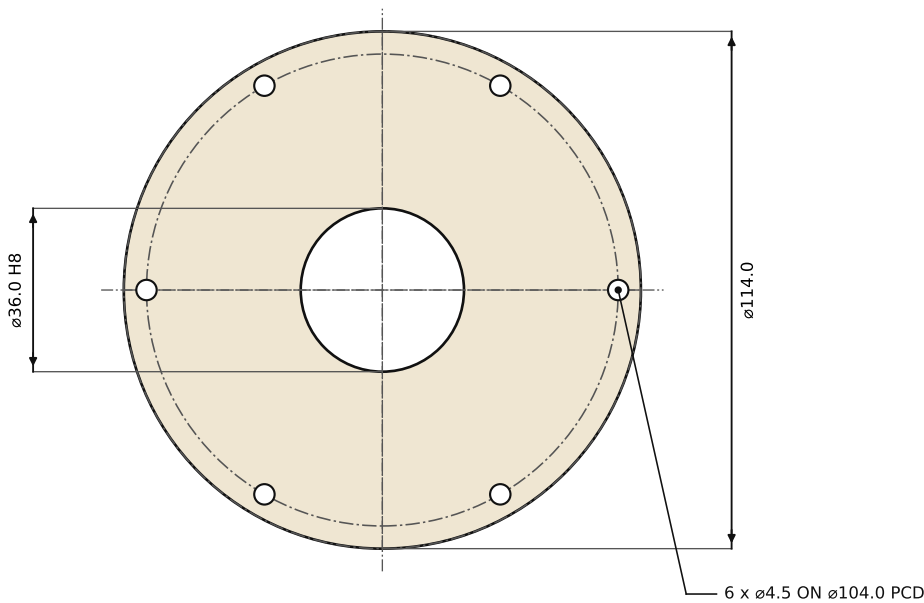
SECTION, BOTH PARTS IN POSITION



ITEM 1 HAND-OFF BOSS, Cu OFHC C101

NOTES

1. The two parts are separate. The 1.0 mm radial gap between them is the thermal choke and must not be filled or shimmed.
2. Item 1 front and rear faces flat within 0.02 and Ra 0.8. The front face is the conduction joint to the anode and the rear face closes the gas plenum in the spreader.
3. Item 1: 9.0 H8 centre bore for the cathode bush. The bore is at anode potential, so the bush is not optional.
4. Gas: 6 x 1.6 from the rear face to within 2.0 of the front face, then 6 x 0.9 x 2.0 metering lands, on a 16.0 PCD. Deburr the bore breakout.
5. Item 2 carries the magnet stack and the casing. Both end faces flat and parallel within 0.03.
6. Item 2 is Macor rather than a conductor on purpose: it is the last thermal barrier between the hand-off path and the rear cap magnets, which are the hottest magnets in the assembly at 72 °C.
7. Masses: item 1 74.7 g, item 2 229.2 g.



ITEM 2 MAGNET SHELF, Macor MGC

MOUNTING PLATE - HAND-OFF BOSS AND MAGNET SHELF

AF-MPDT Breadboard / Rev 3

DRG No.
AFMPDT3-006

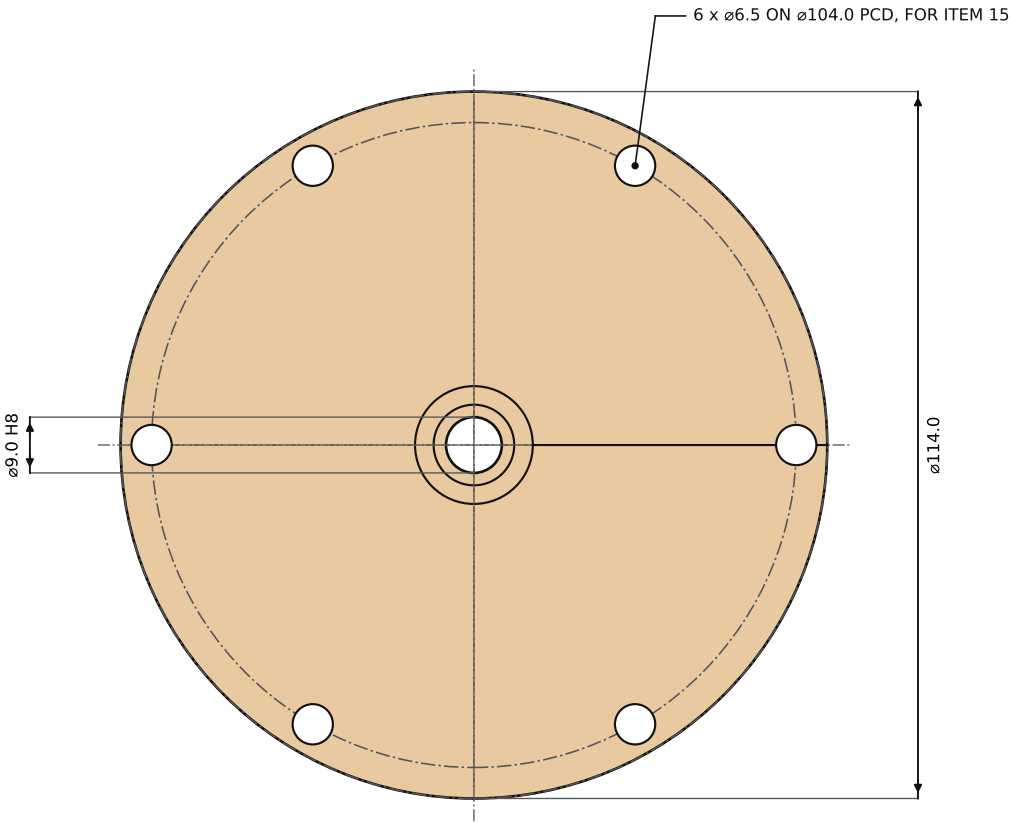
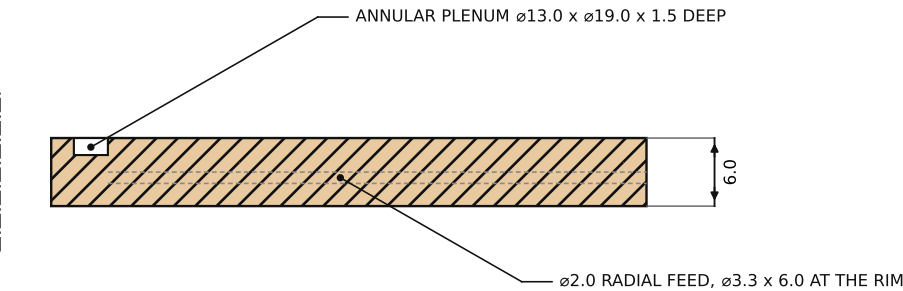
MATERIAL
Cu OFHC C101 / Macor MGC

SCALE
1:1 / 2:1

DATE
2026-09-12

ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED.
REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.

SECTION



FRONT FACE

NOTES

1. This disc is the reason the design works. It takes the heat out to full diameter before the hand-off, turning a 1.1×10^{-3} m² interface into a 1.0×10^{-2} m² one.
2. Both faces flat within 0.03 and Ra 0.8. The front face seals the gas plenum against the hand-off boss, metal to metal.
3. $\phi 9.0$ H8 centre bore for the cathode bush.
4. Gas plenum: $\phi 13.0 \times \phi 19.0 \times 1.5$ deep annular groove in the front face, fed by one $\phi 2.0$ radial passage to the rim, counterbored $\phi 3.3 \times 6.0$ deep to take a 1/8 in feed tube, silver brazed.
5. The six $\phi 6.5$ holes take alumina bushes, item 15, which keep the mounting screws off the copper. The spreader sits at anode potential; the cold plate does not.
6. THE EXTERNAL FEED CONNECTION IS AN OPEN INTERFACE. The gas line and the cathode gland both leave through the rear, and the hand-off face has to stay flat, so the routing has to be settled with whoever designs the cold plate.
7. Mass 531 g.

SPREADER

AF-MPDT Breadboard / Rev 3

DRG No.
AFMPDT3-007

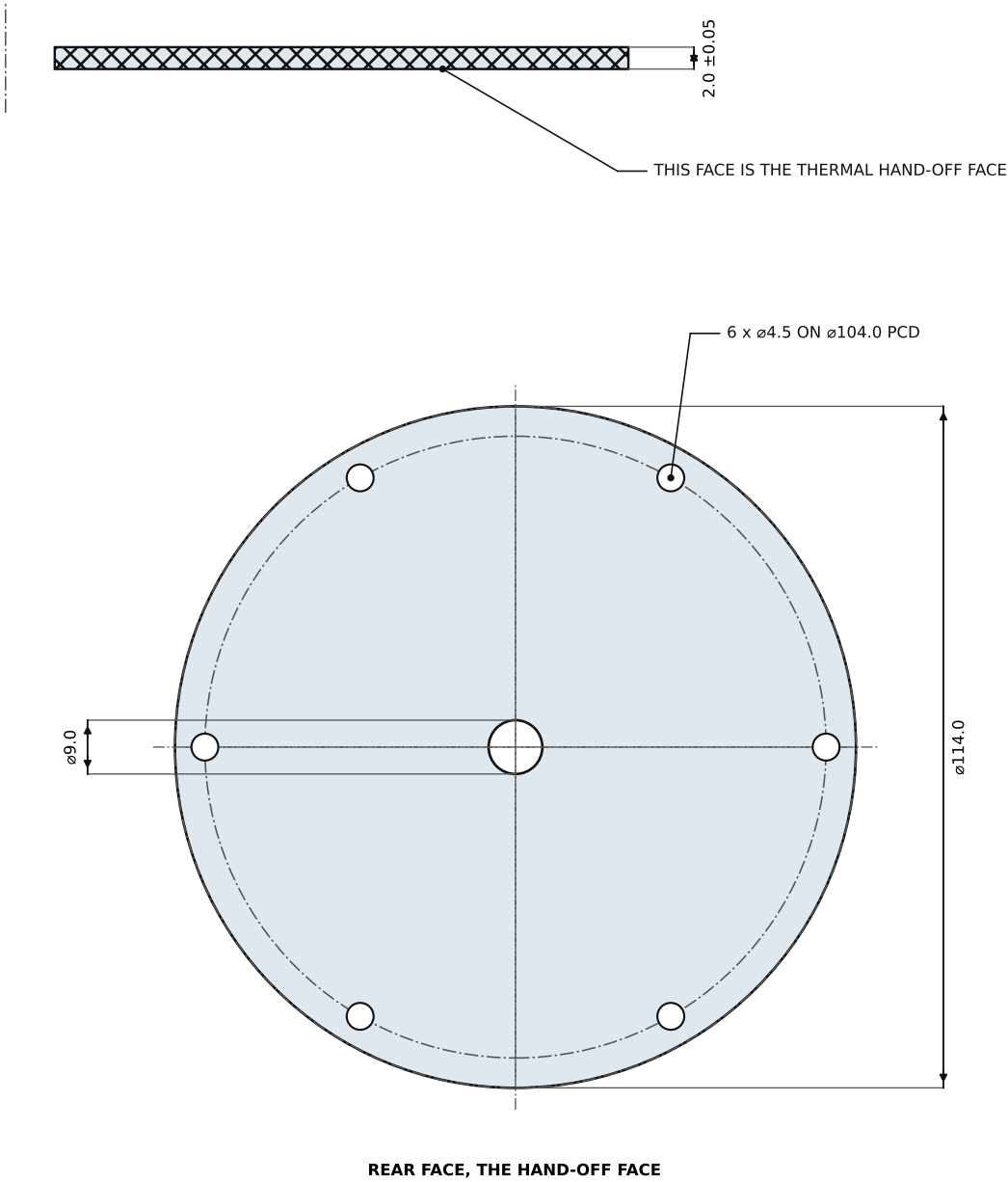
MATERIAL
Cu OFHC C101

SCALE
1:1

DATE
2026-09-12

ALL DIMENSIONS IN mm. GENERAL TOL. ± 0.1 UNLESS STATED.
REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.

SECTION



NOTES

1. Sintered aluminium nitride, thermal conductivity not less than 170 W/(m K), dielectric strength not less than 15 kV/mm.
2. It does two jobs at once. It is the electrical break between the anode potential and the cold plate, and because it runs the full diameter at 170 W/(m K) it also spreads the heat. The second job is why Concept A lost only 67 K to Concept D rather than the hundreds of kelvin the area ratio implies.
3. Rear face flat within 0.05 over the full diameter and Ra 0.8. It is the mounting interface and the thermal hand-off face.
4. The external interface must reach at least 1000 W/(m² K) for the magnets to stay under 100 °C at 500 W. Below about 520 W/(m² K) they pass it. Use a thermal interface material.
5. Do not machine dry. AlN is hard and brittle; grind it.
6. Mass 65.5 g.

ISOLATION PLATE

AF-MPDT Breadboard / Rev 3

DRG No.
AFMPDT3-008

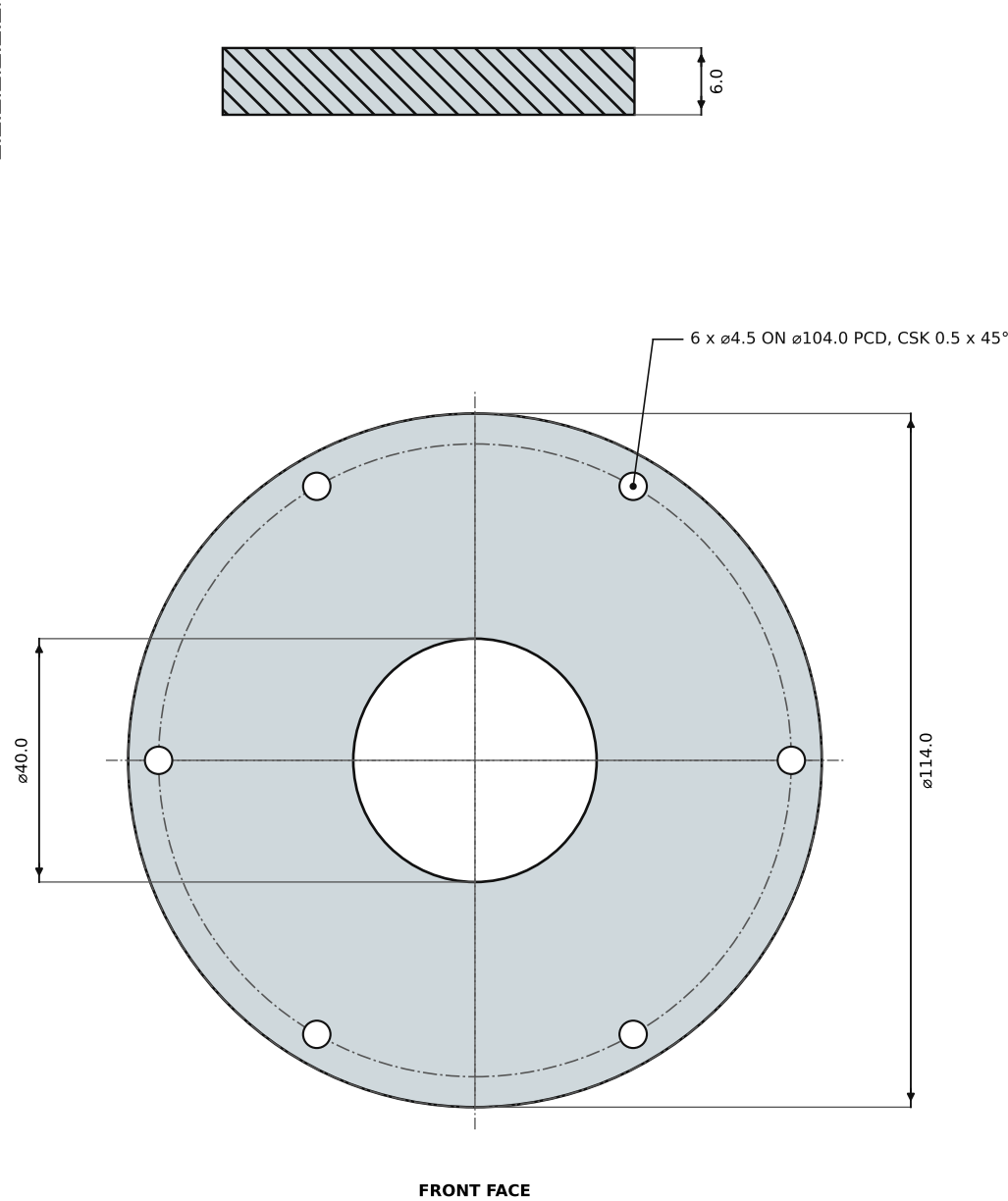
MATERIAL
AlN, sintered aluminium nitride

SCALE
1:1

DATE
2026-09-12

ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED.
REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.

SECTION

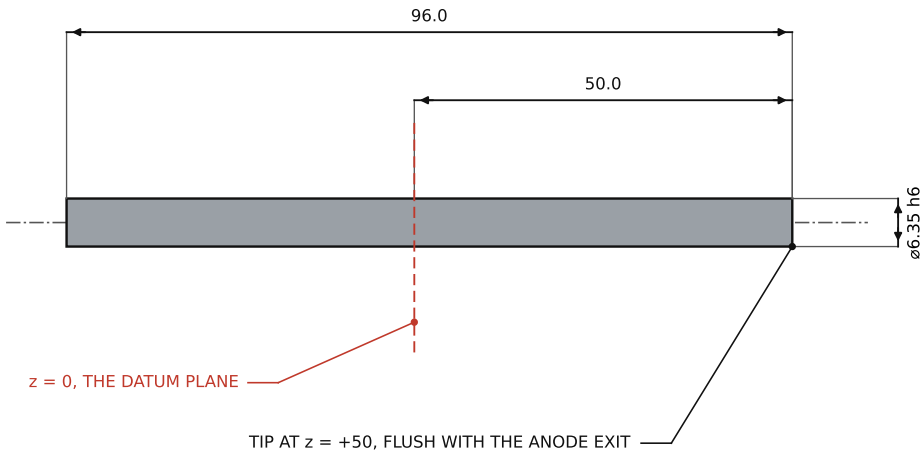


NOTES

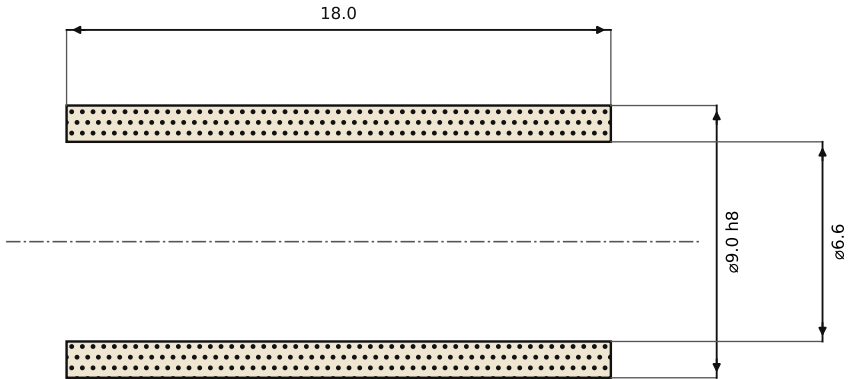
1. KEEP IT AUSTENITIC. A ferromagnetic ring here would short out the front cap flux across the bore and cost most of the field. 304, 316 or 321 are all acceptable; free machining 303 is not, because cold work can raise its permeability.
2. Check with a magnet before fitting. If it attracts, reject it.
3. Faces flat and parallel within 0.05. It spreads the clamp load onto the magnet stack and the casing.
4. 40.0 bore clears the magnet bore by 2.0 on radius.
5. It radiates about 2 W in service, which is why it is in the thermal model at all.
6. Mass 425 g.

FRONT RETAINING RING		
AF-MPDT Breadboard / Rev 3		DRG No. AFMPDT3-009
MATERIAL 304 SS	SCALE 1:1	DATE 2026-09-12
ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED. REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.		

ITEM 13 CATHODE ROD (1:1)



ITEM 14 CATHODE INSULATING BUSH (4:1)



NOTES

- CATHODE, item 13
1. Tungsten with 2 % thoria, or 2 % lanthana. Ground finish.
 2. Tip at z = +50, flush with the anode exit. Set it with a depth gauge: the applied field that performance scales with is defined at the tip, and the field falls steeply behind it.
 3. Record whether the tip is coned or square. It changes arc attachment and it changes run to run comparisons.
 4. Grind the sealing length to Ra 0.4.
 5. If a compression gland is used it must have a PTFE or graphite ferrule. A stainless ferrule will not swage onto tungsten and can chip the rod.
- BUSH, item 14
6. Macor MGC. It insulates the cathode from the copper hand-off boss and the copper spreader, both of which sit at anode potential. 1.2 mm of Macor stands off well over 600 V.
 7. ø6.60 bore gives 0.125 radial clearance on the rod.
 8. The cathode is NOT in the thermal model. It carries its own heat load and its own path to the rear, and including it needs a sheath boundary condition that does not exist yet. The reported temperatures are therefore slightly optimistic.
 9. Masses: rod 58.7 g, bush 1.3 g.

CATHODE ROD AND INSULATING BUSH

AF-MPDT Breadboard / Rev 3		DRG No. AFMPDT3-010
MATERIAL W 2 % ThO2 / Macor MGC	SCALE 1:1 / 4:1	DATE 2026-09-12
ALL DIMENSIONS IN mm. GENERAL TOL. ±0.1 UNLESS STATED. REMOVE ALL BURRS AND BREAK SHARP EDGES 0.3. DO NOT SCALE.		